

# Monetary Policy Transmission to Consumption: Inequalities by Gender and Race

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## **Abstract**

This paper estimates the causal effects of monetary policy shocks on household consumption, with additional analysis of labor market and income responses, disaggregated by gender and race. I find heterogeneity in consumption responses to contractionary monetary policy shocks: after six quarters, households headed by black women experience the largest decline in durable goods spending, while households headed by black men exhibit the largest consumption reallocation toward essential non-durable goods and services. These gaps persist after accounting for differences in household education, debt, and income, but are partly explained by differences in marital status and spousal insurance against shocks. The analysis provides estimates of consumption-to-income response ratios by group and shows that contractionary, rather than expansionary, shocks drive aggregate consumption responses. These findings highlight the importance of accounting for intersectional demographic heterogeneity in evaluating the distributional effects of monetary policy.

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# 1 Introduction

Understanding the distributional effects of monetary policy is crucial for effective policy-making. Previous studies have emphasized that the effects of monetary shocks on household outcomes differ by wealth, income, or labor market outcomes (Auclert 2019; Cloyne et al. 2020; Coibion et al. 2017). Others explore the disproportionate burden of contractionary policy on certain racial groups through labor market and wealth outcomes (Seguino and Heintz 2012; Bartscher et al. 2022; Amberg et al. 2022; Long 2026). Responses may differ across individuals with different racial and gender identities because these identities are associated with differences in exposure to labor market shocks, household income and wealth composition, debt burdens, and the ability of other household members to adjust labor supply and insure consumption. However, no papers have estimated the effects of monetary policy on consumption responses by gender and race. It is therefore unclear whether consumption inequality can be fully understood by analyzing groups by education and finances, or whether other breakdowns of groups can shed light on other mechanisms that determine responses.

This paper addresses this question by documenting the transmission of monetary policy to consumption separately for white men, white women, black men, and black women. It is the first, to my knowledge, that estimates and explains consumption responses by gender and race with a state-of-the-art methodology and shock identification method. Contractionary policy shocks decrease consumption, raise unemployment, and decrease income for groups at different rates. These differences may reflect not only differences in income losses, but also differences in household structure, income composition, debt exposure, and the ability to adjust labor supply in response to a shock. Studying heterogeneous reactions solely through income inequality therefore obscures additional reasons why monetary policy affects groups differently.

The analysis employs several data sources. I use the Consumer Expenditure Survey to measure household consumption and income and the Current Population Survey to measure labor market outcomes. I use the monetary policy surprise series by Bauer and Swanson (2023), which provides a recent high frequency measure that is relevant and exogenous. This series measures high-frequency revisions to the expected path of monetary policy around

FOMC announcements. I refer to these measured innovations as shocks hereafter. The use of high frequency shocks contributes to the literature on gender and racial inequality.

I estimate the effects of monetary policy shocks from 1988 to 2019 using the local projections instrumental variables methodology (Jordà et al. 2020). Local projections have been widely used to estimate the effects of monetary policy (Coibion et al. 2017; Bauer and Swanson 2023). I also conduct several exercises that confirm the robustness of the results including using alternative monetary policy shock series, controls, and time periods.

This paper first estimates how monetary policy affects consumption across U.S. households disaggregated by gender and race. Because racial and gender groups are not homogeneous, I examine race and gender jointly rather than separately. This intersectional approach reveals heterogeneity that can be obscured when households are aggregated along either dimension alone. The results show that contractionary monetary policy shocks lead to larger declines in consumption among black households than white households, with the largest short-run response concentrated in durable goods. Following a 25 basis point (bp) contractionary shock, durable goods consumption falls by 7.5% among households headed by black women, compared with 4.7% among households headed by white men after six quarters. In the longer term, 12 quarters after the shock, I estimate a relatively smaller but sustained decrease in total spending among households headed by black men and white women.

I next examine how households reallocate expenditures across categories following a monetary policy shock. At the same six-quarter horizon, households headed by black women exhibit little change in spending on non-durable goods and services. However, other households increase such spending by 1.25% for households headed by black men and 0.21% for households headed by white men. More broadly, following a contractionary shock, consumption falls for durables and non-essential goods while rising for essential non-durables and services. These results provide new evidence on how households substitute expenditures across categories following a monetary policy shock.

Racial and gender differences in consumption responses persist when accounting for mechanisms commonly used in the literature such as education, mortgage debt, and income. Instead, household structure is an important dimension for understanding the transmission of monetary policy shocks. Married households exhibit durable consumption responses that

are smaller in magnitude than those of comparable single households, consistent with an additional insurance margin against shocks provided by the labor supply of a spouse. I find little systematic racial or gender heterogeneity in household consumption responses by marital status. By jointly examining consumption, income, and labor market responses across marital, racial, and gender groups, this paper highlights household structure as an important dimension of monetary policy transmission heterogeneity and provides evidence on the role of spousal insurance in consumption smoothing.

Finally, I contribute two estimates that can inform quantitative models studying optimal monetary policy. The first is the household consumption-to-income response ratio to a monetary policy shock. The ratio for spending on durables is larger than that for non-durables and services. The second estimate is the difference in the response of consumption to contractionary versus expansionary shocks. Contractionary, rather than expansionary, shocks drive overall household consumption responses.

## 1.1 Related Literature

This paper contributes to the growing literature studying how monetary policy affects inequality (see surveys by Seguino (2019) and Kappes (2023)). Important channels of the transmission of monetary policy to consumption include earnings heterogeneity, income composition, debt burden, and uncertainty (Coibion et al. 2017; Auclert 2019; Amberg et al. 2022; Grigoli and Sandri 2026). Empirical evidence finds that income effects dominate substitution effects on consumption and savings decisions following a change in interest rates (Coibion et al. 2017).

My paper provides new causal estimates of the effects of monetary policy shocks on household consumption, labor market outcomes, and consumption-income ratios disaggregated by both gender and race. To date, only one study—Albert and Gomez-Fernandez (2024)—has examined consumption responses by race, with no prior work considering the joint role of gender and race.

This paper advances the literature in several key ways. First, I estimate consumption responses by gender within racial groups, introducing a new dimension of heterogeneity and highlighting the role of household composition in monetary policy transmission. Second,

the use of a state-of-the-art identification strategy and instrumental variables framework enables me to identify differential consumption and labor income responses by race and gender—specifically, that contractionary shocks reduce consumption and salary income more for black than white households—effects not detected in Albert and Gomez-Fernandez (2024). Third, I decompose changes in non-durable goods and services, documenting how households reallocate spending in response to shocks. Fourth, I estimate total household income responses and show that income inequality by race rises following contractionary shocks. Finally, I complement existing findings by showing that financial income and savings exhibit greater volatility for black households following shocks. Together, these contributions provide a more detailed and causally credible understanding of the distributional effects of monetary policy. I relate my findings to the following channels of monetary policy transmission.

*The earnings heterogeneity channel* traces the effects of monetary policy shocks on consumption through labor market outcomes and earnings. In the US, contractionary policy tends to raise unemployment more for black than white individuals, and disproportionately affects women, who are more concentrated in low-wage, precarious jobs due to historical barriers in education and employment (Thorbecke 2001; Rodgers 2008; Seguino and Heintz 2012; Ume and Williams 2019; Bergman et al. 2022; Bartscher et al. 2022; Bennani 2023). Black and women employees are also more likely to be last hired and first fired (Seguino and Heintz 2012).

Gendered labor market responses also reflect sectoral composition: men are more concentrated in cyclical industries, making them more exposed to downturns (Duzhak 2021; Flamini et al. 2023).<sup>1</sup> This can trigger the “added earner effect,” where other household members—often women—enter the labor force in response to male job loss (Juhn and Potter 2007). Racial and gendered norms also shape labor supply as white women are more likely to exit the labor force for unpaid care, while black women often continue working alongside community caregiving responsibilities (Banks 2020).

These dynamics matter for monetary transmission. Married households can partially

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<sup>1</sup>There is contrasting evidence on how monetary policies impact gender gaps in the labor market in cross-country studies. Flamini et al. (2023) find that contractionary policy negatively affects men more than women, Takhtamanova and Sierminska (2009) find no significant relationship in the OECD, and Braunstein and Heintz (2008) show that women’s employment is more negatively impacted in developing countries.

insure against income shocks through joint labor supply adjustments, smoothing consumption more than single households (Browning et al. 2014; Ortigueira and Siassi 2013; Wu and Krueger 2021). Because black women are more often single heads of household, with both paid and unpaid work responsibilities, contractionary policy can reduce their consumption more sharply. I contribute to the literature by providing new evidence that spousal labor supply adjustments through the added earner effect dampen the transmission of monetary policy to household consumption.

*The income composition channel* explains household consumption responses through fluctuations in different sources of income. Montecino and Epstein (2017) and Bartscher et al. (2022), among others, find that changes in household wealth following monetary policy shocks outweigh other effects on employment and mortgage refinancing. Albert and Gomez-Fernandez (2024) attribute racial gaps in consumption to disparities in wealth. However, most studies examine gender and racial inequalities in employment and wealth separately. This paper contributes by linking consumption, labor market, and income responses across household types.

Business and financial income are not theoretically supposed to react differently for demographic groups. What matters, are the size of the share of each category for total household income. Since assets compose a larger share of incomes for white than black households, financial incomes of white households are likely more exposed to monetary policy (Bartscher et al. 2022; Long 2026).

*The debt burden channel* highlights how higher interest rates from contractionary monetary policy increase household debt burdens, particularly through mortgage costs. Cloyne et al. (2020), Wong (2015), and Koeniger et al. (2022) estimate that households with mortgages drive the aggregate household consumption response to monetary policy. Black households are disproportionately affected due to higher initial mortgage rates, limited refinancing access, and higher insurance and property tax burdens (Aronowitz et al. 2020; Avenancio-León and Howard 2022; Gerardi et al. 2023).

Households that rent may be affected by changes in rental payments following contractionary policy. As higher mortgage rates reduce buying, rental demand rises, potentially increasing rents (Dias and Duarte 2022). Since a larger share of black households rent, and

rental payments often do not contribute to credit histories, black households will experience a magnified effect of monetary policy (Puig 2022).

This paper also adds to the literature on the marginal propensity to consume (MPC) and the possible state dependence of impulse response functions. My estimated consumption-income response ratios complement MPCs out of unemployment and transfer shocks by race and gender (Ganong et al. 2020; Patterson 2023; Puig 2026). The analysis speaks to asymmetries in monetary policy effects, contrasting studies that find either equal distributional impacts or progressive effects favoring black households (Furceri et al. 2018; Lee et al. 2021; Nakajima 2023; McKay and Wolf 2023). Although expansionary policies are progressive as in Del Canto et al. (2025), my findings demonstrate that economists must consider the sizable losses from contractionary policies.

## 2 Data & Methodology

### 2.1 Data

The time frame of analysis is 1988-2019. The data for consumption and income are U.S. household survey data from the Consumer Expenditure Survey (CE). The CE surveys households every quarter about their previous three months of consumption, income, and expenditures. It is a rotating panel, meaning that households are dropped once they are surveyed for four consecutive quarters. I impute data prior to 2004 following Coibion et al. (2017) to account for the CE’s imputation of data after 2004. The data are aggregated to a quarterly frequency and adjusted using a four-quarter moving average. Values are deflated by the Consumer Price Index to create a real series in 2019 prices and normalized by effective household size.<sup>2</sup> See Appendix A for additional details on the CE data cleaning procedure.

Categorization of durable goods, non-durable goods and services, and total income follows the classifications in Coibion et al. (2017). Total income is composed of salary, business,

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<sup>2</sup>The constructed total consumption CE series is highly correlated (0.93) with the quarterly Real Personal Consumption Expenditures series by the Bureau of Economic Analysis. The constructed total after tax income CE series is correlated (0.55) with the Income After Taxes series by the Bureau of Labor Statistics (BLS). The lower correlation between the income series can be attributed to differences in their construction. The constructed CE series is quarterly and excludes certain households, while the BLS series is annual, includes all households, and needs to be converted to real values.

financial, and other income sources such as transfers net of taxes.<sup>3</sup> Capital gains and losses, including dividends, are included in the definition of household financial income. The CE has less comprehensive income and wealth data relative to other surveys; however, it is still preferable given this paper’s focus on consumption outcomes.

To calculate the gender and racial gaps in outcomes, households in the CE are disaggregated by the demographics of their head of household. The CE assigns a reference person for each household who provides the most information on family composition, income, taxes, and expenditures. The researchers classify this individual as the head of household due to their extensive knowledge of household details and role in the survey. In households with spouses, the CE allows for either person to be the reference person. This flexibility is an advantage over other popularly used household surveys that automatically assign this position to the male in the household. Details on the age, education, sex, and race of the reference person and other family members are provided in the raw data. I follow Cloyne et al. (2020) and Coibion et al. (2017) in using this classification of reference person as the household head. The racial analysis focuses on white and black households due to the small sample sizes of Asian, Native American, and Hispanic households.<sup>4</sup>

The analysis of labor market outcomes uses data from the Current Population Survey (CPS) rather than the CE to obtain more precise estimates of the effects of monetary policy shocks. Ideally, labor market responses would be measured at the individual level using the same data as the consumption analysis. However, the CE interview collects income at the household level and employment status is reported retrospectively over the preceding 12 months. The number of unemployed household heads in the CE is relatively small, limiting estimate precision. In contrast, the CPS provides a large, nationally representative sample with monthly individual-level observations, allowing for more accurate estimation of labor market responses across the demographic groups of interest. I construct quarterly averages of the unemployment rate, labor force participation rate, and employment-population ratio using CPS adult final weights. The quarterly series are adjusted using a four-quarter moving

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<sup>3</sup>I do not estimate transfers separately from total income by gender and race due to data limitations.

<sup>4</sup>The CE collects data on Hispanic households starting in 2009, with data on a few Latin American nationalities starting in 2003. This limits studying this ethnicity with the entire sample and comparing responses to white and black populations.

average to follow the adjustment procedure used for the CE data.

Quarterly observation counts are stable over the sample period. Appendix Table A.1 shows average quarterly unweighted cell counts, weighted population shares, and effective sample sizes by demographic group of interest. Appendix Table A.2 shows that the majority of observations report positive expenditure. Since the CE accounts for attrition through their sample rotation collection procedure, subgroup composition is stable over time. The standard deviation of the weighted share of households headed by white men is 0.1, white women is 0.09, black men 0.01, and black women 0.01. Within subgroups, this standard deviation is smaller.

The monetary policy shocks used are the high frequency series by Bauer and Swanson (2023) at a quarterly frequency. Following Bauer and Swanson (2023), this series is used in conjunction with controls for industrial production sourced from the Board of Governors of the Federal Reserve System, the consumer price index (CPI) from the BLS, and the excess bond premium from Favara et al. (2016). The control variables are measured as the average value in each quarter.

## **2.2 Descriptive Statistics**

The CE data is consistent with the gender and racial composition of the US. In the sample, 46% of households are headed by women. Also, 87% are white and 13% are black households, mirroring the division of race within the U.S. population during this time period. Table 1 shows that average consumption is considerably lower and more volatile for black than white households. Income and savings are lower for black (women) versus white (men) households. Household differences in consumption and income are generally larger by race than gender. Black individuals, especially black men, also have the highest unemployment rates. Labor force participation is higher for men than women, but higher for black women than white women.

This distribution reflects the trends in the US population, as men and white individuals have more access to high paying and stable jobs. It may also be explained by the proportion of dual income earner households in each group since black, especially women, heads of household are more likely to be single and thus support their family alone both in income

Table 1: Median quarterly outcome variable over the full sample

| Outcome                                             | White<br>Men | White<br>Women | Black<br>Men | Black<br>Women |
|-----------------------------------------------------|--------------|----------------|--------------|----------------|
| HH consumption of durable goods (\$)                | 687          | 596            | 491          | 382            |
| HH consumption of non-durable goods & services (\$) | 3,742        | 3,554          | 3,005        | 2,589          |
| HH total (after tax) income (\$)                    | 29,218       | 22,512         | 20,820       | 13,482         |
| HH labor income (\$)                                | 22,486       | 12,920         | 16,225       | 7,453          |
| HH business income (\$)                             | 2,647        | 1,556          | 1,104        | 428            |
| HH financial income (\$)                            | 3,520        | 2,980          | 1,619        | 907            |
| HH savings (\$)                                     | 2,255        | 1,855          | 577          | 311            |
| Unemployment rate (%)                               | 5.4          | 5.0            | 11.7         | 10.2           |
| Labor force participation rate (%)                  | 72.7         | 57.6           | 65.8         | 59.5           |

Note: The table shows median quarterly outcomes for white men and women and black men and women. Consumption and income are at the household level (HH = Household), identified by the head of household in the CE data. Household amounts are in 2019\$ and normalized by effective household size. Savings statistics describe households that have positive savings since most households have zero savings. Labor market statistics are at the individual level from the CPS.

and care work. Black households also have less access to public and private goods; relying more on community care that is often provided by black women (Banks 2020).

In households headed by men, 59% of wives who do not work in formal employment instead take care of the household, providing unpaid care work and increasing familial welfare. In households headed by women, husbands who do not to work are usually retired (64%) or ill or disabled (20%), possibly providing pensions to the household, but not explicitly contributing to housework. This arrangement may lead to an additional burden on working women heads who provide unpaid care to their elderly, ill, or disabled husbands. This inequality of within-household distribution of income and unpaid work likely affects decision-making around spending on consumption (Doss 2021).

The significant differences in distribution of work and income between households disaggregated by gender and race puts into question the generalizability of aggregate household results. It is thus important to analyze the effects separately by groups of households.

## 2.3 Monetary Policy Shocks

Interest rates are endogenous to economic variables, as central bankers often set policy in response to aggregate economic fluctuations. It is therefore crucial to correctly identify monetary policy shocks to estimate the effects of unexpected policy rate changes on economic outcomes. Numerous methods of identifying monetary policy shocks have been used in the literature. High-frequency movements in interest rates around FOMC announcements capture revisions to the expected path of policy rather than only the target rate decision itself. These surprise revisions affect asset prices, borrowing costs, and household balance sheets even when the policy action is widely anticipated (Nakamura and Steinsson 2018; Bauer and Swanson 2023).

The monetary policy shocks used in this paper are the conventional high-frequency series of Bauer and Swanson (2023). This series is aggregated to a quarterly frequency by the sum of all shocks within the quarter.<sup>5</sup> The Bauer and Swanson (2023) series measures intraday innovations in interest-rate expectations from 1988-2019 and thus isolates policy-induced changes in financial conditions, including forward guidance and communication about future rates. The series captures surprise changes in expected future policy rates at announcement windows, even when the policy move itself is anticipated. The relevance of the series is higher relative to other measures due to the authors expanding the set of monetary policy announcement events with publicly available data.

## 2.4 Empirical Methodology

The effects of monetary policy shocks on various demographic groups are estimated through local projections as in Jordà et al. (2020). Local projections are performed with the shock series as an instrumental variable to estimate the causal effect of monetary policy (Ramey 2016). I follow Bauer and Swanson (2023) in regressing each outcome of interest on the 2-year Treasury yield instrumented with the high-frequency monetary policy surprise measure at each horizon.<sup>6</sup> The following equation is estimated:

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<sup>5</sup>See the robustness section for a discussion of alternative monetary policy shock series. The results are consistent across different measures.

<sup>6</sup>I use the end-of-quarter two-year Treasury yield. The measure of the yield at the end of the period is suggested by Bauer and Swanson (2023) as the most adequate for their shock instrument.

$$x_{g,t+h} = \alpha_{g,h} + \psi_{g,h}(L)Z_{t-1} + \beta_{g,h}Y_t^{2y} + \varepsilon_{g,t+h} \quad \text{for } h = 0, \dots, H-1, \quad (1)$$

where  $x$  is the variable of interest,  $\alpha$  is a constant,  $\psi_{g,h}(L)$  is a polynomial in the lag operator of degree 1 (allowing for 2 lags),  $Z$  is a vector of control variables,  $\beta$  is the coefficient of interest,  $Y^{2y}$  is the two-year Treasury yield, and  $\varepsilon$  is the residual. The vector  $Z$  includes lags of the variable of interest, two-year Treasury yield, log industrial production, log CPI, and the excess bond premium.<sup>7</sup> Regressions are run separately for each demographic group  $g$  (individuals or households headed by white men, white women, black men, and black women). All variables are over quarterly time  $t$  through to time horizon  $h$ . As a benchmark, the estimates cover  $H = 20$  quarters.

Equation 1 is estimated with a GMM estimator using the Bauer and Swanson (2023) shock as the instrument for  $Y^{2y}$ .<sup>8</sup> Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors are used. In all estimations, the first-stage F-statistic of the instrumental variable is above the Stock and Watson (2012) rule of thumb of 10 for a relevant instrument (see statistics reported in Appendix B).

### 3 Monetary Policy Effects on Aggregates and Consumption

The local projections produce conventionally signed effects on most macroeconomic aggregates. The first row of Figure 1 shows that a 25bp contractionary monetary policy shock leads to a decrease in industrial production of 1.76%, increase in unemployment of 0.47 percentage points (pp), and decrease in household income of 2.36% relative to their respective counterfactual levels 12 quarters after the shock. The responses of other macroeconomic aggregates are consistent in direction and magnitude with the results of Bauer and Swanson

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<sup>7</sup>Control variables are included to account for variations in macroeconomic outcomes over time as in Gertler and Karadi (2015) and Bauer and Swanson (2023). Log industrial production is included for a measure of economic output, log CPI for overall market price fluctuations, and the excess bond premium for investor sentiment in the corporate bond market.

<sup>8</sup>The Bauer and Swanson (2023) conventional shock is used in this specification to ensure instrument validity. Bauer and Swanson (2023) propose an orthogonalized shock, but the version of this shock that is a valid instrument in a local projections specification is not yet publicly released. See Bauer and Swanson (2023) for a discussion of instrument relevance and exogeneity conditions for each shock. Bauer and Swanson (2023) show that their orthogonalized measure produces estimates of larger magnitude and statistical significance than their conventional measure. The estimates in this paper are thus a lower bound of results.

(2023) estimated via local projections and the same shock series (see Appendix C).

Figure 1: Effects of monetary policy shocks on aggregate variables



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on aggregate variables. Time (horizontal axis) is in quarters. Dark and light grey shaded areas represent one standard deviation and 90% confidence intervals respectively. Subfigure (e) plots the gap in consumption on different types of goods: non-durable goods and services (NDS) minus durable goods (D). Data in subfigure (a) is sourced from the Board of Governors of the Federal Reserve System, (b) from the CPS, and (c)-(e) from the CE.

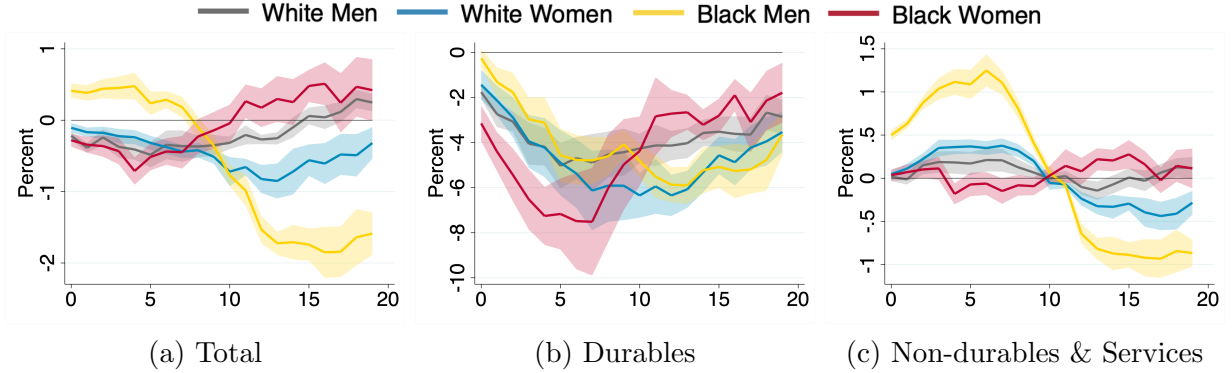
Contractionary monetary policy shocks also lower household consumption, as shown in the second row of Figure 1. A 25bp contractionary shock decreases total consumption by 0.44% and 0.49% at 6 and 12 quarters after the shock, respectively. The consumption of durable goods falls more than that of non-durable goods and services, consistent with the consumption literature (Coibion et al. (2017), among others). This leads to a widening of the consumption gap between non-durable goods and services and durable goods by 4.81% (4.33%) after 6 (12) quarters.

### 3.1 Consumption Responses

Monetary policy shocks impact consumption differently for households headed by black men, black women, white men, and white women. Figure 2 shows that following contractionary

shocks, consumption falls most for black households and least for households headed by white men at various shock horizons.<sup>9</sup> The differences in group responses are statistically significant, see a formal assessment in Appendix D. Since black households on average spend less than white households, as seen in Table 1, this result implies that consumption inequality rises after a contractionary shock.

Figure 2: Effects of monetary policy shocks on household consumption



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption, separately by households headed by white men, white women, black men, and black women. Sub-figures show consumption of (a) total goods, (b) durable goods, and (c) non-durable goods and services. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

Black households experience the largest changes in consumption following monetary policy shocks, although the timing differs across groups. Figure 2a shows that six quarters after a 25bp contractionary monetary policy shock, total consumption decreases by approximately 0.4% for all groups except households headed by black men, whose consumption increases by 0.3%. Over the medium and longer term, however, households headed by black men exhibit the largest decline, reaching 1.5% after 12 quarters, compared with 0.8% for white women, 0.2% for white men, and approximately zero for black women.

Consumption responses also differ by gender. Households headed by black women reduce consumption initially but recover relatively quickly, whereas households headed by white women experience a more persistent decline, reaching a peak response 12 quarters after the shock and recovering more slowly than white men.

<sup>9</sup>For ease of presentation, figures henceforth display one standard deviation confidence intervals. See Appendix Figure D.1 for 90% confidence intervals for Figure 2.

These differences extend to durable and non-durable goods and services. Six quarters after the contractionary shock, households headed by black women reduce durable goods consumption by 7.5%, the largest decline across groups, while non-durable goods and services consumption falls by 0.06%. Durable goods spending initially decreases across groups but remains below -5% for households headed by white women and black men in later quarters, rather than recovering. Spending on non-durable goods and services initially increases for all groups except black women, by 1.25% for black men, 0.35% for white women, and 0.21% for white men at quarter 6. By quarter 12, however, non-durable consumption declines by 0.64% for black men and 0.23% for white women, with these responses remaining persistent thereafter.

### 3.2 Drivers of Consumption Response Gaps

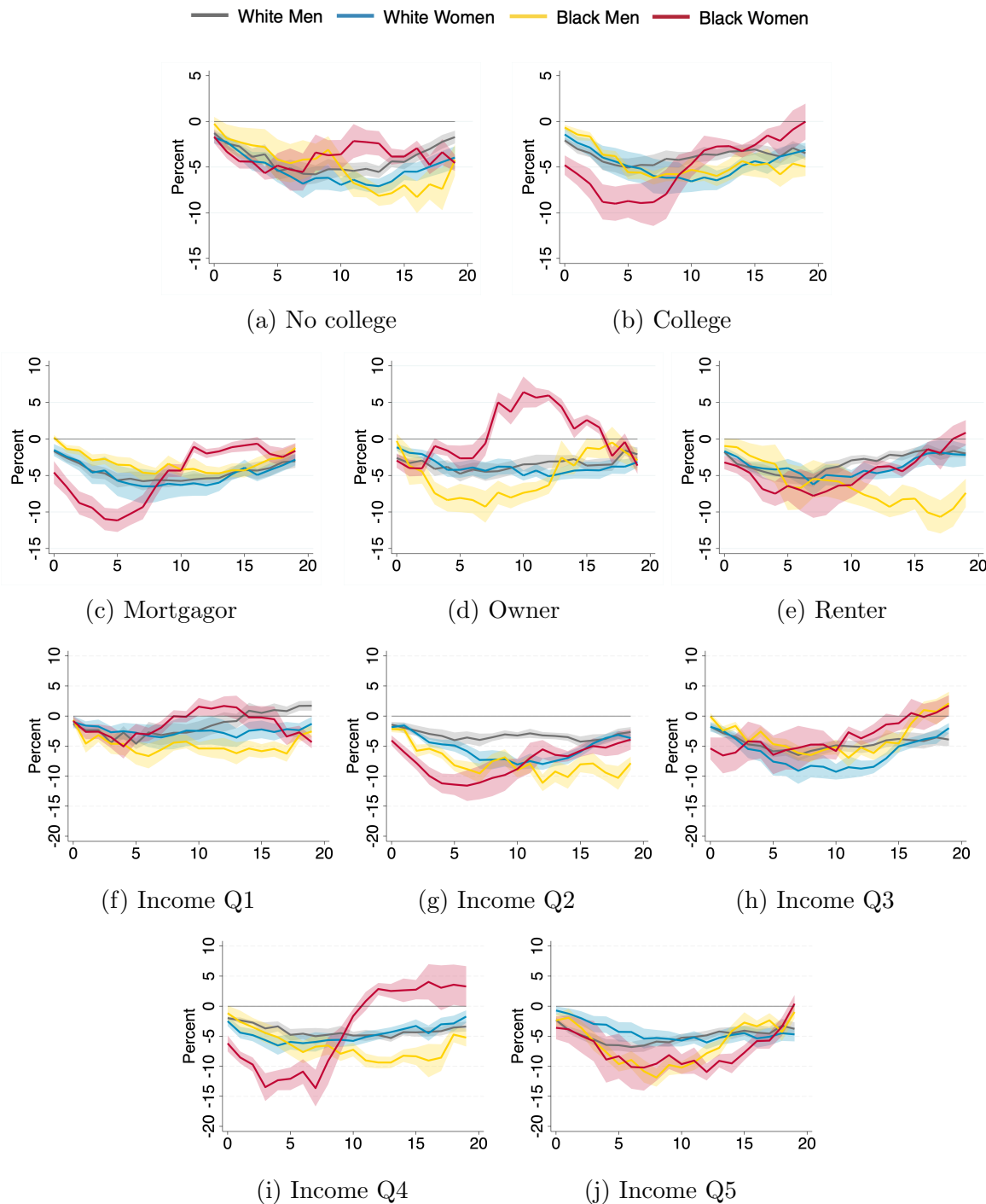
I next investigate whether the gaps in responses in Figure 2 are driven by household characteristics other than gender and race. The analysis in this section focuses on durable goods consumption, as these responses are quantitatively large and account for most of the decline in total household consumption. The following section demonstrates that the conclusions regarding the underlying mechanisms extend to household consumption of non-durable goods and services.

Mechanisms such as education, mortgage debt, income, and age have been proposed to explain heterogeneous responses to monetary policy. If, for example, gender and racial consumption gaps were solely driven by education inequality, these gaps would not exist within subgroups by educational attainment. However, I find that these mechanisms neither fully explain the variation in consumption response to contractionary policy, nor the gender and racial differences in responses. Figure 3 shows that durable goods spending continues to be decreased more by black than white households regardless of these mechanisms.<sup>10</sup>

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<sup>10</sup>This result is consistent with the racial consumption gaps found in Ganong et al. (2020).

Figure 3: Effects of monetary policy shocks on household durable goods consumption by education and finances



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption of durable goods, separately by households headed by white men, white women, black men, and black women. Each row displays different breakdowns of the total sample. Sub-figures compare households whose heads (a) do not have a college degree vs (b) have a college degree, (c) mortgage vs (d) own vs (e) rent their home, and (f-j) quintiles of the family income distribution. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

I first estimate consumption responses to monetary policy separately for households by their head’s educational attainment. The durables consumption of all households without a college degree responds more to contractionary shocks than of households with a college degree. However, Figure 3 shows that racial gaps are prominent among households with college education. Among the college-educated, consumption falls by almost twice the magnitude for households headed by black women than others in the short run. Additionally, households headed by black men and white women have persistently lower consumption responses than other groups regardless of educational attainment in the longer run.

Household mortgage debt also does not explain the gender and racial consumption gaps from Figure 2. To measure the debt burden channel of monetary policy, I separate households by whether they have a mortgage on their house, own their house, or rent. Households with mortgages decrease their spending the most compared to owners and renters in the short run, consistent with Cloyne et al. (2020). However, racial and gender gaps in consumption responses persist within these debt categories. Households headed by black women drive the short run fall in durable goods spending among mortgagors in Figure 3. Also, owner and renter households headed by black men decrease their durables consumption more than white households in the medium and long term, respectively. The debt burden channel therefore also restricts the consumption of black renter households.

Gender and racial consumption gaps additionally persist within different levels of family income.<sup>11</sup> In the first year after a contractionary shock, the response of durables is larger for households in the lowest income quintile compared to the second, third, and fourth quintile among households headed by white men. However, this is not always the case for other demographic groups. Black households reduce their durable spending to a greater degree than white households at all ranges of the income distribution within the first six quarters after a shock. For example, the consumption of households headed by black women falls by over 10% for those in the second, fourth, and fifth quintiles of household income.

The age of the head of household additionally does not drive gender and racial consumption gaps (see estimates in Appendix E). There is no clear trend between younger versus

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<sup>11</sup>Family income percentile thresholds are the same for all gender and racial groups. These results hold when using a measure of household labor earnings instead of total income. I also find similar gaps when disaggregating households by their savings and wealth. Estimates are available upon request.

older households as in Leahy and Thapar (2022) or Wong (2015). Instead, consumption gaps in Figure 2 are maintained within several age bands, consistent with Cloyne et al. (2020). In the first six quarters, households headed by black women decrease their durable goods consumption more than other groups among many age bands.

Geographical location is also important to consider, as monetary policy’s effects on inequality vary by the share of each demographic group in that region (Seguino and Heintz 2012). In Appendix F, I estimate that consumption reacts more to monetary shocks for black than white households regardless of location, especially in the short term. I also find that households headed by black women decrease their durable goods consumption with a higher magnitude in rural areas and in places that are most exposed to shocks.

These results demonstrate that monetary policy does not influence consumption to the same degree for all demographic groups. Additionally, gender and racial consumption gaps are not fully understood with conventional mechanisms for monetary policy transmission. To further explore the contribution of the analysis by household gender and race, I conduct two variance decomposition exercises of consumption. The first examines the unconditional contribution of gender and race to household consumption, while the second estimates this contribution to consumption conditional on the response to monetary policy shocks. I find that gender and race does matter for understanding the conditional response of consumption to monetary policy shocks.

Analysis by gender and race can explain part of the variation in household consumption further than the variation explained by household finances. As evidence of the additional contribution of gender and race, I first decompose the variation in consumption into its common (aggregate) and group-specific components.<sup>12</sup> I find that gender and race controls explain over 2% of the variance unexplained by the aggregate component. Income is the largest contributor to the unexplained variance from the aggregate component; explaining about 13%. College education and mortgagor status each contribute 5%, while marital status and age contribute less than 1% to the unexplained variance.

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<sup>12</sup>The variance decomposition is estimated in a two-stage regression of households over quarters of time. In the first stage, the aggregate component is calculated by regressing the log of consumption on time dummies. The second stage regresses the residuals from the first stage on each group control. The variation is given by the  $R^2$  estimates from each regression.

The second variance decomposition exercise is an analysis of which mechanisms explain the variation in the consumption response to monetary policy shocks for all households. The variation in the response of all households is decomposed into components explained by each mechanism of household heterogeneity.<sup>13</sup> I find that four quarters after the shock, race and gender explain 75.3% of the variance of all households, income explains 67.8%, marital status explains 21%, mortgagor status or age explain 15%, and education explains 5.6%. Race and gender continue to explain a large share of variation in quarter 6, second only to income.<sup>14</sup> It is therefore important to estimate consumption responses to monetary policy shocks separately by groups to fully understand the variation in responses.

## 4 Consumption Reallocation

The response of non-durable goods and services to monetary shocks reveals an important dimension of household adjustment that is obscured by aggregate consumption. Although total consumption declines following a contractionary monetary policy shock, most of this decline is driven by the sharp contraction in durable expenditures. The responses of non-durable goods and services are relatively smaller, but display substantial heterogeneity across demographic groups both in magnitude and sign. Figure 2c shows that households headed by black men, white men, and white women significantly increase non-durable expenditures in the short run, whereas households headed by black women reduce them. These results suggest that many households partially offset declines in durable consumption by reallocating expenditures toward non-durable goods and services.

I separately estimate the responses of non-durable goods and services subcategories to examine which ones account for this reallocation. Table 2 shows the response of each subcategory of non-durables and services spending to a 25bp contractionary shock. I focus on the 6-quarter horizon, where the average response of households headed by black men reaches its

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<sup>13</sup>The total variance in the response of durables is computed from equation 1 as in Figure 1. The effect of monetary policy shocks on consumption is next estimated separately for each household group: race and gender; education; mortgagor status; income percentile; marital status; and age. The total variance of all households is then decomposed into the portion explained by the response of each demographic group.

<sup>14</sup>Since these demographic groups are correlated, the sum of variances is larger than the total variance for all households. When households are separated by race and gender in addition to each group, a considerably larger share of variance is explained. This suggests that the pooled estimate of the response of consumption for all households likely understates the variance in consumption responses between groups.

peak of 1.25%.<sup>15</sup> Households headed by white men and women decrease these expenditures by 0.21% and 0.35% respectively. On average, households allocate 60% of expenditures to non-durable goods and 40% to services. Food is the largest expenditure category, comprising 40% of all non-durables and services spending. The significant increase in food spending among all households in Table 2 is thus a large driver of the overall rise in non-durables and services following contractionary shocks.

Table 2: Effect of monetary policy shocks on categories of household non-durable goods and services consumption

|                           | White<br>Men   | White<br>Women | Black<br>Men   | Black<br>Women |
|---------------------------|----------------|----------------|----------------|----------------|
| Non-durables & Services   | 0.21 (0.13)*   | 0.35 (0.08)**  | 1.25 (0.19)**  | -0.06 (0.13)   |
| Non-durables              | 0.28 (0.11)**  | 0.52 (0.15)**  | 1.73 (0.30)**  | 0.37 (0.17)**  |
| - Clothing, footwear      | 0.30 (0.18)*   | 1.16 (0.17)**  | 1.30 (0.41)**  | 0.27 (0.59)    |
| - Food                    | 0.36 (0.10)**  | 0.64 (0.14)**  | 1.68 (0.34)**  | 0.73 (0.17)**  |
| - House furnishings       | -0.97 (0.33)** | 0.55 (0.31)**  | 2.27 (1.15)**  | -2.25 (0.97)** |
| - Tech., entert.          | -0.29 (0.17)** | -0.36 (0.22)** | 0.44 (0.69)    | -0.57 (0.35)*  |
| - Transportation          | -0.07 (0.27)   | -0.81 (0.32)** | 2.27 (0.42)**  | -0.48 (0.54)   |
| Services                  | 0.38 (0.22)**  | 0.16 (0.11)*   | 0.67 (0.29)**  | -0.75 (0.35)** |
| - Care                    | 0.61 (0.30)**  | 0.97 (0.21)**  | 1.59 (0.73)**  | -1.69 (0.34)** |
| - Utilities, maintenance  | 1.14 (0.16)**  | 0.37 (0.22)**  | 2.08 (0.36)**  | 0.33 (0.36)    |
| - Insurance               | 0.15 (0.13)*   | -0.11 (0.33)   | 1.27 (0.89)*   | -2.10 (0.47)** |
| - Tech., ed., and entert. | 0.45 (0.49)    | 1.02 (0.31)**  | 1.26 (0.55)**  | -1.98 (0.40)** |
| - Transportation          | -1.81 (0.58)** | -0.73 (0.24)** | -0.84 (0.36)** | -0.83 (0.30)** |

Note: The table shows the percent response of each household expenditure category to a 25bp contractionary monetary shock at quarter 6. Category labels are shortened for presentation: the fifth row of house furnishings includes other house non-durable goods; the sixth row is technology and entertainment non-durable goods; the ninth row includes child, elder, house, and personal care services; the tenth row is house utilities and maintenance services; and the twelfth row is technology, education, and entertainment services. Newey–West HAC standard errors in parentheses. \*  $p < 0.32$ , \*\*  $p < 0.10$ . Data used is sourced from the CE.

The category-specific estimates in Table 2 suggest that households shift their consumption away from interest-sensitive and postponable purchases toward more essential goods and services. Across most demographic groups, spending on technology and entertainment goods

<sup>15</sup>The 6-quarter horizon is additionally shown to focus on the peak difference in responses between demographic groups. The estimates in Table 2 are consistent in the short run at alternative horizons. See Appendix G for estimates at a 3-quarter horizon.

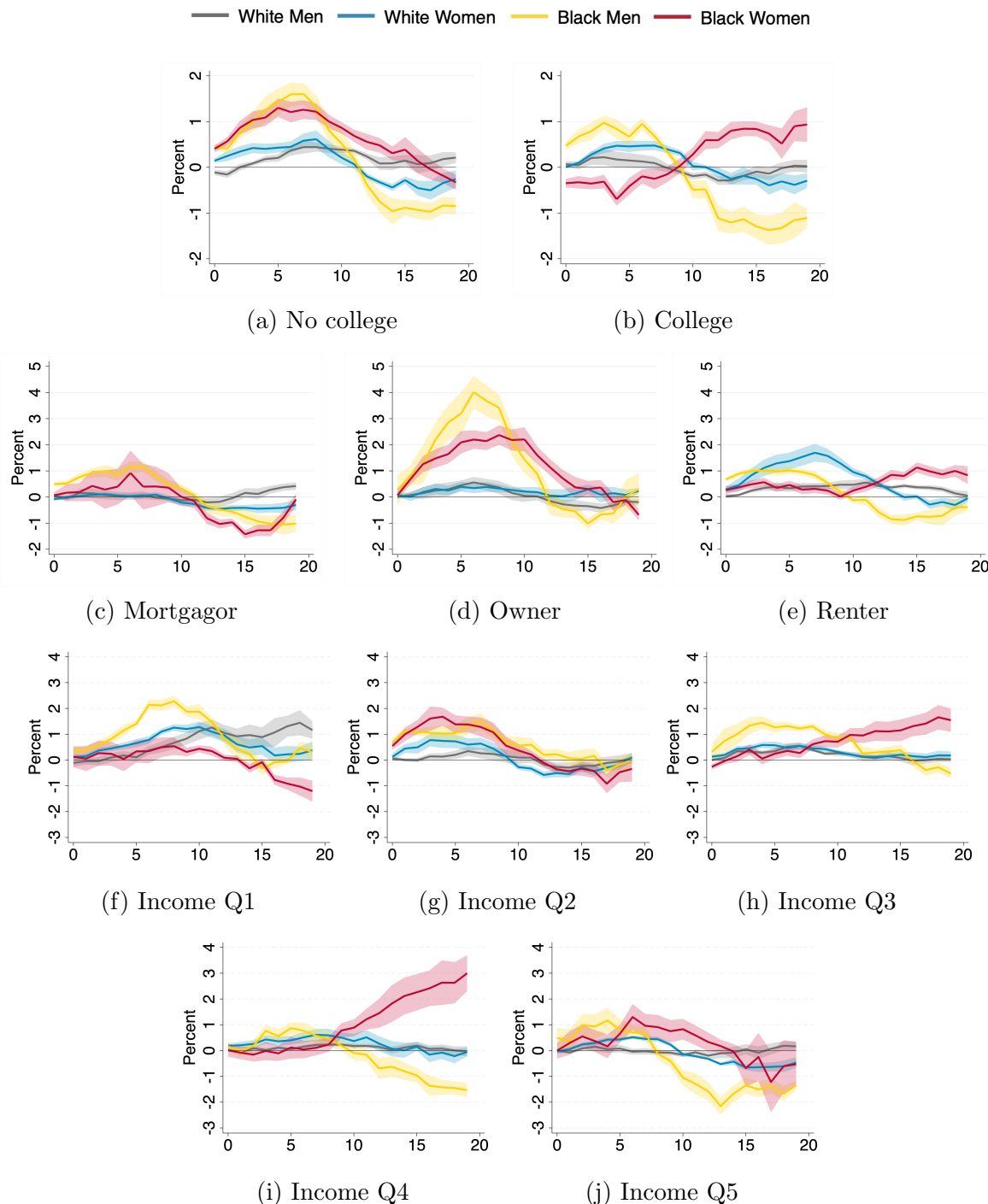
and transportation is significantly reduced, while expenditures on clothing, food, and services such as utilities, maintenance, technology, and entertainment is significantly increased. These responses are consistent with households postponing discretionary purchases while boosting smaller-scale expenses that are used on a daily basis in the home.

There are also important differences by household gender and race in which categories of non-durables and services are reduced. Households headed by black women decrease their consumption of non-durables and services at a statistically significantly higher magnitude than other households (formal test in Appendix Table D.1). This is especially true for technology, entertainment, and education. However, household headed by black women also decrease their spending on household furnishings, insurance, and household care services while other households increase spending on these goods and services. Combining the response in Table 2 with those of durable goods, it is clear that households headed by black women experience the largest overall contraction in consumption among the four demographic groups in the short run.

#### **4.1 Mechanisms Underlying Consumption Reallocation**

I next investigate whether heterogeneity in consumption reallocation is explained by conventional monetary policy transmission mechanisms. Figure 4 shows that gender and racial differences in non-durable goods and services responses remain across household head education, homeownership status, and income quintile. The subgroup analysis compares three mechanisms underlying the observed reallocation that operate at different margins: category-specific price effects, precautionary stockpiling, and liquidity constraints.

Figure 4: Effects of monetary policy shocks on household non-durable goods and services consumption by subgroups



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption of non-durable goods and services, separately by households headed by white men, white women, black men, and black women. Each row displays different breakdowns of the total sample. Sub-figures compare households whose heads are (a) do not have a college degree vs (b) have a college degree, (c) mortgage vs (d) own vs (e) rent their home, and (f-j) quintiles of the family income distribution. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

While a price channel can explain the decline in total and durable consumption following contractionary monetary policy shocks, it cannot fully account for consumption reallocation or the demographic heterogeneity in spending responses. The sharp contraction in durable expenditures shown in Figure 2b and the larger decline among mortgagors relative to homeowners and renters shown in Figure 3 over the first six quarters are both consistent with this mechanism. Higher interest rates raise the cost of financing durable purchases, particularly vehicles and home-related investments, leading households that are more exposed to interest-rate changes to reduce these expenditures. However, the price channel does not explain which households reallocate spending toward non-durable goods and services. Figure 4 shows that mortgagors do not exhibit the strongest increases in non-durable and services expenditures following a contractionary shock. Instead, this reallocation is more pronounced among homeowners and renters in the short term. Moreover, the demographic differences persist within housing debt groups, suggesting that additional mechanisms contribute to the observed heterogeneity in consumption responses.

A second potential mechanism is precautionary stockpiling. If households anticipate that prices will rise following a monetary policy shock, they may increase storable goods purchases, leading to temporary increases in food, clothing, and other household goods. The observed increases in food and clothing expenditures are consistent with this mechanism. However, stockpiling cannot account for the broader pattern of responses, as expenditures also increase in certain services (home, care, education, and entertainment) that are not easily stored for future consumption. The estimates therefore suggest that anticipated price increases alone cannot explain consumption reallocation.

The cross-group evidence instead points most strongly to a liquidity-constraint or financial-frictions channel governing the household ability to reallocate consumption. Reallocation from durables toward non-durables in the short term is pervasive among renters, homeowners without mortgages, lower-income and non-college degree households, and all white households. In contrast, mortgagors and households headed by black women, especially those with college degrees and lower income, exhibit little evidence of such mitigating reallocation. Within the first six quarters after a shock, these households reduce durable expenditures, while non-durable spending remains flat or also declines, leading to a larger over-

all decrease in consumption. This non-reallocating group of households experiences higher financial frictions on average limiting their liquidity, which can reduce their ability to substitute across consumption categories in response to contractionary policy shocks. Overall, liquidity and balance-sheet constraints appear to be key determinants of whether households can exploit the consumption reallocation margin following contractionary monetary policy shocks.

Since conventional monetary policy transmission mechanisms do not fully explain the variation in consumption responses, I next explore other mechanisms that could drive the differences in consumption responses by gender and race. The next section investigates whether household labor market adjustment and income responses to monetary policy account for why some households are less able to smooth consumption following contractionary monetary policy shocks. Black households are more likely to be credit constrained, and thus are expected to decrease their spending most due to disproportionate job and income losses.

## 5 Household Composition and Consumption Insurance

In this section, I examine how household composition shapes the transmission of monetary policy to consumption through labor market adjustment and intra-household risk sharing. Household composition is particularly relevant because liquidity constraints and income fluctuations may be absorbed differently in single or two-adult households. While the consumption response of a single household more closely reflects the preferences and labor market outcomes of one individual, the response of a married household reflects joint decision-making given shared responsibilities or divisions of paid and unpaid work. Comparing single and married households therefore provides evidence on whether intra-household insurance mitigates the effects of contractionary monetary policy.

Because marital status is likely correlated with household characteristics that also affect consumption dynamics, I first reweight single households to match the observable characteristics of married households before comparing the two groups.<sup>16</sup> I then estimate consumption

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<sup>16</sup>Single households are reweighted to match the observable characteristics of married households within each sex-race group and year in the sample. The reweighting accounts for a polynomial in age and indicators for college education, employment status, mortgagor status, region, and quarter using family weights. Additionally, households whose marital status changes during the sample period are omitted from the estimation

responses to contractionary shocks separately for the reweighted sample of single households and for married households. This nonparametric reweighting approach reduces differences in observable characteristics between the two groups, allowing the comparison to focus on differences associated with household composition rather than observable demographic and socioeconomic characteristics.

Figure 5 shows married and adjusted single household consumption responses to contractionary monetary policy shocks (see Appendix Figure H.1 for estimates for non-adjusted single households).<sup>17</sup> Married households reduce their durable consumption to shocks to a lesser degree than comparable single households. Estimates in Appendix Table H.1 show that the married-single difference is statistically significant for households headed by white men and black women over the first twelve quarters and for black men over the first six quarters. These results suggest that marriage is an additional mechanism enabling consumption smoothing that is unavailable to single households. Responses of nondurable expenditures are more heterogeneous and do not exhibit a systematic pattern across household composition, suggesting that the primary effect of spousal insurance operates through postponement of durable purchases (see Appendix Figure H.1).

Within marital status groups, demographic groups respond similarly to contractionary monetary policy. In Appendix Table H.2, I find no statistically significant evidence of systematic gender or racial differences in durable spending responses within married or single household groups. There are, however, two exceptions: differences between households headed by white men and black men are statistically significant at several horizons, as are differences among married white households at longer horizons. Thus, while marital status is associated with differences in consumption responses, the data provide little evidence of broad demographic heterogeneity in responses conditional on marital status.

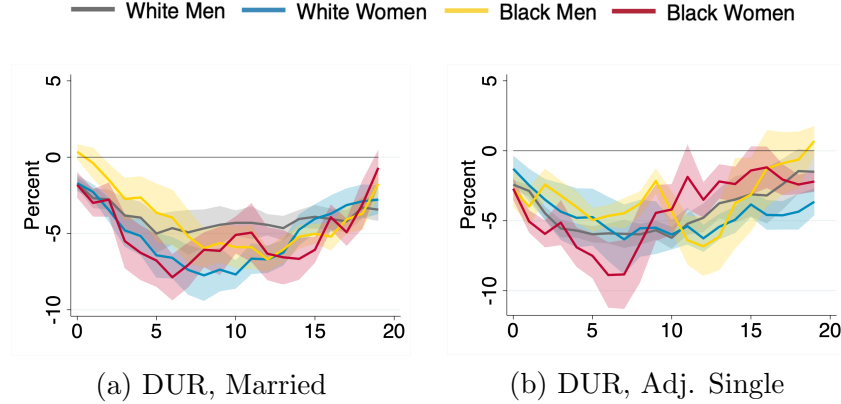
These findings suggest that marital status provides an additional margin of consumption insurance following aggregate income shocks. One potential source of this insurance is the

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(see observation counts in Appendix Table A.1 .

<sup>17</sup>The results by marriage status are consistent with households separated by the number of income earners. Earners are either the head of the household or the spouse. Households who have one (two) income earner have similar responses as those who are single (married). However, income earners fluctuate within households and the CE interviews households for short periods. The results by income earners are therefore more difficult to interpret than those by marital status, which more often stays constant over time.

Figure 5: Effects of monetary policy shocks on household consumption by marital status

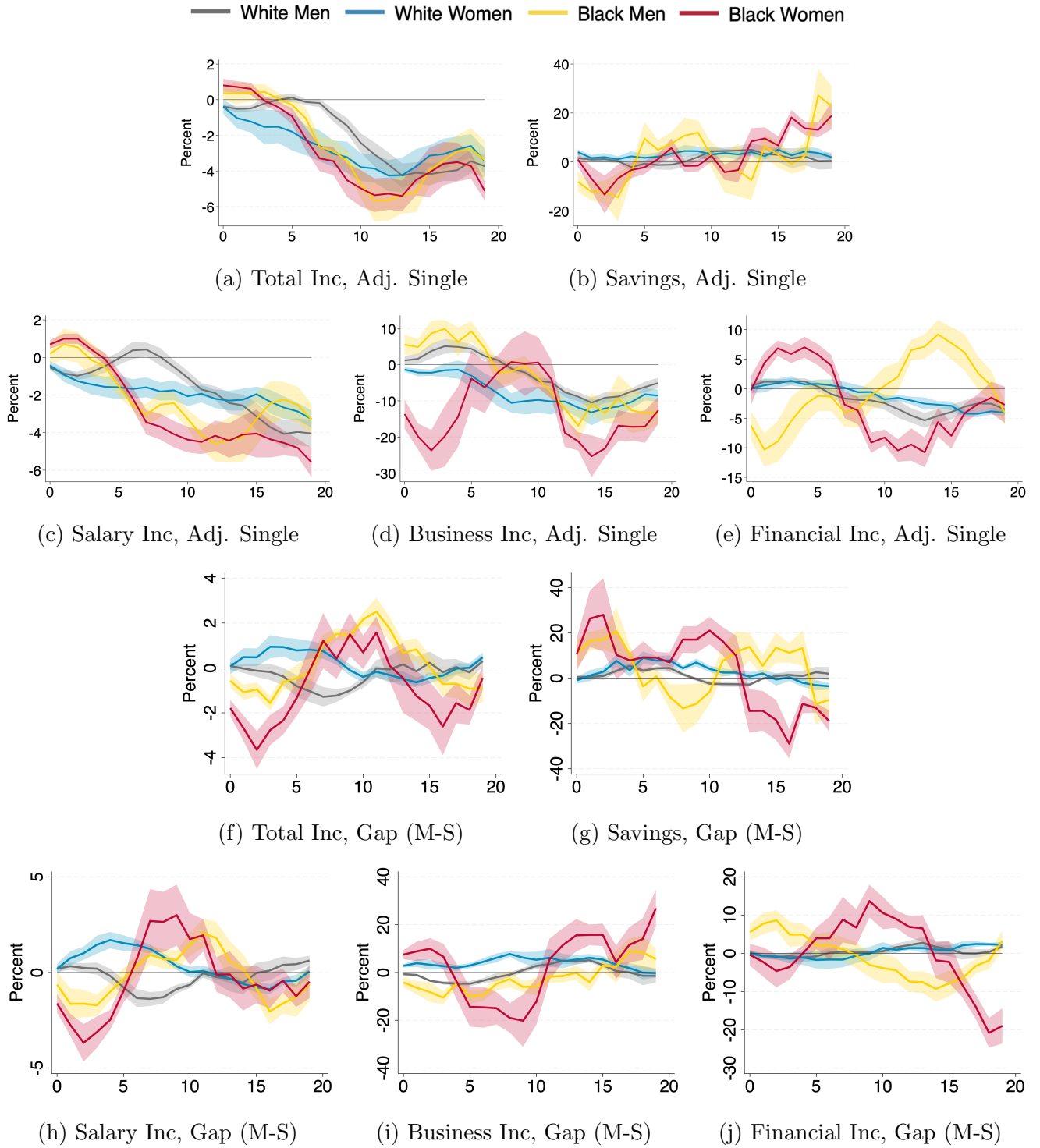


Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption of durable goods (DUR) separately by households headed by white men, white women, black men, and black women. Sub-figures compare households that are married and households that are single adjusted with weights to match married household characteristics. Non-adjusted single household responses and marital status gaps are displayed in Appendix Figure H.1. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

ability of a spouse to adjust labor supply in response to a negative household income shock. Unlike single households, married households can therefore partially offset an income loss through the labor supply of the other spouse.

I find evidence of spousal insurance against income fluctuations following monetary policy shocks. The first two rows of Figure 6 show that a 25bp contractionary shock reduces total income, salary income, and business income for most adjusted single households within the first 12 quarters after a shock. The last two rows show that income gaps between married and single households also change following a contractionary shock. The dynamics of the total and salary income gaps, shown in Figures 6 f and h, are similar: during the first year following the shock, the married-single income gap narrows among black households but begins to widen approximately six quarters after the shock. These estimates suggest that while married household salary income declines as labor market shocks affect multiple household members, married households experience a faster and more persistent recovery than single households. The married-single income gap also widens for households headed by white women six quarters after the shock.

Figure 6: Effects of monetary policy shocks on household finances by marital status



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household income and savings, separately by households headed by white men, white women, black men, and black women. Total income is divided into salary, business, and financial income. Sub-figures a-e show estimates for single households adjusted with weights to match married household characteristics. Sub-figures f-j show the gap between married and adjusted single households. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

The total and salary income patterns are consistent with theories of spousal insurance, which predict that household income fluctuates less than individual spousal income due to labor supply adjustments partially offsetting adverse income shocks (Browning et al. 2014). Since contractionary shocks decrease total incomes less for married than single households in the medium run, it follows that their consumption also falls less than that of single households.

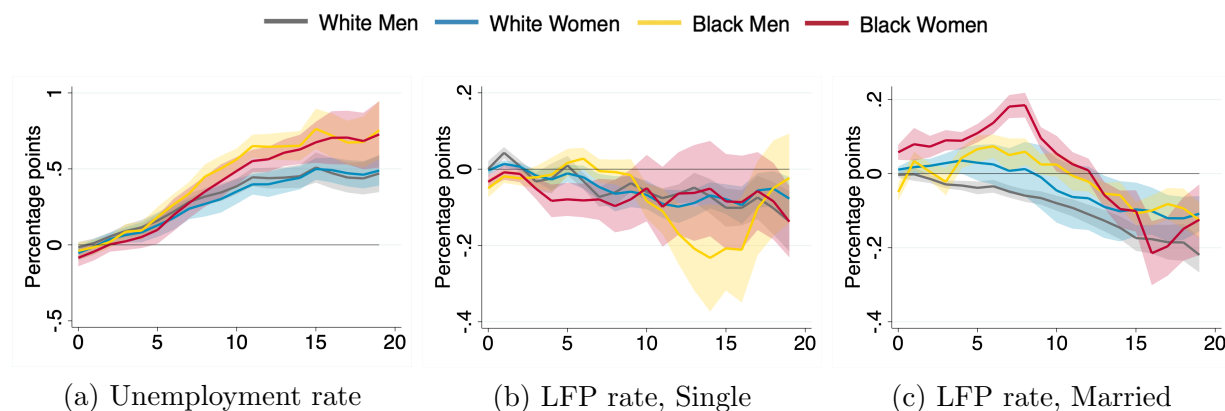
Gains in non-labor income can also compensate for labor market losses and smooth household consumption. Business incomes rise for single men households, but fall for single women households in the first year after a contractionary shock. For most household groups, financial incomes rise within the first year after the shock and then decline. While savings balances are steady or positive in the medium run for most households, they also fall for black households within the first year. Married-single savings and financial income gaps widen in the first year among black households and also among white households in the medium and longer term. The fact that business and financial incomes comprise a larger share of total income for white than black households can explain the racial differences in consumption responses. Higher financial income and savings also aid white and married households in smoothing consumption alongside falls in salary incomes.

The breakdown of income sources in Figure 6 shows that the composition of household income can influence the response of consumption to monetary shocks. However, the responses of all financial measures to contractionary shocks fluctuate more for black than white households at most horizons. My estimates therefore suggest that reductions in consumption due to contractionary shocks are driven by falling incomes and savings.

In Appendix I, Figures a-b examine household salary income conditional on at least one spouse being employed. While single household salary income continues to fall more than for married households, it decreases by less than 5% for all groups in the short and longer term. This is a substantially smaller response than that of the full sample, suggesting that monetary policy affects labor market outcomes largely through the extensive margin of employment rather than solely through adjustments in earnings among those who remain employed. I next examine this mechanism directly using individual labor market outcomes from the CPS.

I estimate the effect of monetary shocks on individual labor market outcomes to further disentangle the differences in income responses by marital status. Figure 7a shows that the unemployment rate rises for all groups following a 25bp contractionary monetary policy shock; by 0.2pp after 6 quarters and by 0.4-0.6pp after 12 quarters. The racial gap in the unemployment rate widens by 0.13pp after 12 quarters (see Appendix I Figure c for estimates of the gap). The widening of the racial unemployment gap is consistent with the widening of the racial salary income gap documented in Figure 6c.

Figure 7: Effects of monetary policy shocks on individual labor market outcomes



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on individual labor market outcomes, separately by white men, white women, black men, and black women. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CPS.

In Appendix J, I investigate whether occupational concentration drives racial differences in unemployment responses. I estimate that monetary policy shocks decrease employment in industry more than in services in the medium and long term.<sup>18</sup> However, racial and gender gaps remain within the responses of many sectors. Sector exposure therefore does not fully explain the demographic gaps. The component of the unemployment response that

<sup>18</sup>This estimation is motivated by Duzhak (2021), who finds that occupational concentration drives racial differences in unemployment following downturns. The relatively larger impact on industry is consistent with Flamini et al. (2023). I find that changes in industry employment are driven by the construction sector. Changes in services employment are driven by utilities, wholesale and retail trade, and real estate sectors. Black employee concentration in wholesale and retail trade and service jobs and non-white employee concentration in construction could drive the racial gap in unemployment responses. Men's concentration in industry jobs could drive gender gaps. However, the first row in Appendix Table J.1 shows that black employment falls more in construction than white employment following a contractionary shock.

is unexplained by occupational concentration calls for further investigation into labor market inequalities.

Alongside changes in unemployment, individuals adjust their labor force participation (LFP). Figure 7b shows that LFP falls for all single individuals by 0.02-0.08pp and 0.06-0.17pp after 6 and 12 quarters, respectively, following a 25bp contractionary shock. The LFP for married white men also falls in the short and long term in Figure 7c. However, the LFP for other married individuals rises in the short term; for example by 0.14pp for black women after 6 quarters. This result supports the added income earner effect that married individuals enter the labor force to support their households during times of unemployment. I confirm that individuals who enter the labor force become employed rather than unemployed in Appendix I Figures d-e.<sup>19</sup> Black income earners experience higher unemployment rates following contractionary shocks, and thus their spouses have higher LFP in this period. Therefore, the responses of labor market outcomes to monetary policy shocks also support the theory of spousal insurance.

### 5.1 Decomposition of the Monetary Policy Added Earner Effect

The empirical evidence points to spousal insurance through an added earner as a mechanism that may explain differences in household responses to monetary policy shocks by marital status. The previous section documents three facts that are jointly consistent with this mechanism: in the medium run, (1) married households experience smaller income declines, (2) married households experience smaller consumption declines, and (3) married women increase labor force participation following contractionary monetary policy shocks. Although these facts are drawn from different data sources, together they suggest that married households can partially insure against income shocks through adjustments in spousal labor supply. Household composition may therefore play an important role in shaping the transmission of monetary policy shocks by determining household ability to smooth income and consumption.

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<sup>19</sup>I estimate the impact of a 25bp contractionary shock on the ratio of employment to the total population and find that this employment ratio generally falls following a contractionary shock. However, the employment ratio rises for married women, by 0.15pp for black women after 6 quarters, shown in Appendix I. This is almost the same magnitude as the rise in the LFP rate for this group in Figure 7c.

I illustrate this mechanism by analyzing a collective household framework with endogenous labor supply as in Browning et al. (2014), Blundell et al. (2016), and Wu and Krueger (2021), embedded in a Heterogeneous Agent New Keynesian environment. A household composed of spouses  $j \in \{1, 2\}$  chooses consumption  $c$  and labor supply  $h_j$  to maximize joint lifetime utility (see Appendix K for details). Following Kaplan et al. (2018), the transmission of a monetary policy shock  $\epsilon$  to household consumption operates through the real interest rate  $r$  and household income; in this case, the primary earner's wage. Consistent with the empirical estimates in section 5, the pass-through of the income shock is governed by wage exposure ( $\gamma$ ), asset-based self-insurance ( $\pi$ ), and spousal labor insurance ( $\eta$ ):

$$\Delta c_{i,t} = \sigma \Delta r(\epsilon_{i,t}) + (1 - \pi_{i,t} - \eta_{i,t}) \Delta \gamma(\epsilon_{i,t}),$$

where the spousal insurance term is defined as the income share multiplied by the Frisch elasticity of labor supply for the secondary earner:  $\eta = \frac{w_2 h_2}{w_1 h_1 + w_2 h_2} \phi_2$ . For single households,  $h_2 = 0$  implies  $\eta = 0$ , so consumption smoothing relies solely on asset buffers. For married households, added income earners mitigate the pass-through of primary-earner wage shocks.

The model motivates a reduced-form measure of the added earner insurance channel by relating empirical estimates of spouses' labor supply responses to the consumption smoothing advantage of marriage. For each demographic group  $g$ , I define the following ratio:

$$\xi_g = \frac{\sum_h \Delta L_{g,h}^{Spouse}}{\sum_h [\Delta c_{g,h}^M - \Delta c_{g,h}^S]},$$

where  $\Delta L^{Spouse}$  is the spouse's LFP response and  $\Delta c^M$  and  $\Delta c^S$  are the consumption responses of married and adjusted single households, respectively, to the contractionary monetary policy shock at each horizon  $h$ . The ratio  $\xi$  measures the cumulative spouse labor supply response per percentage point of consumption insurance associated with marriage. A larger positive  $\xi$  indicates greater reliance on the added earner margin to smooth consumption following a contractionary monetary policy shock.

There is substantial group heterogeneity in the relationship between the consumption smoothing associated with marriage and the LFP response of spouses. Table 3 reports cu-

mulative responses through six quarters following a 25bp contractionary shock.<sup>20</sup> Among households headed by men, the difference in cumulative consumption responses between married and adjusted single households is positive and statistically significant for both white and black households, indicating that married households experience smaller cumulative consumption declines. At the same time, female spouses increase LFP following the contractionary shock, with a substantially larger response among black women than white women (0.64pp versus 0.17pp). The resulting added earner ratio is larger for black men than white men ( $\xi = 0.04$  versus  $\xi = 0.01$ ). The relatively large labor force response of black women spouses suggests that labor supply is an important margin through which black households insure consumption against monetary policy shocks, likely due to relatively lower liquid assets compared to white households.

Table 3: The added income earner insurance ratio to monetary policy shocks

| Group       | Consumption Difference<br>$\Delta c_M - \Delta c_S$ | LFP Spouse<br>$\Delta L_{spouse}$ | Added Earner Ratio<br>$\xi$ |
|-------------|-----------------------------------------------------|-----------------------------------|-----------------------------|
| White men   | 12.28 (8.56)*                                       | 0.17 (0.09)*                      | 0.01 (0.01)*                |
| White women | 0.65 (9.78)                                         | -0.15 (0.04)*                     | -0.23 (3.47)                |
| Black men   | 16.15 (10.84)*                                      | 0.64 (0.08)*                      | 0.04 (0.03)*                |
| Black women | 20.53 (14.14)*                                      | 0.15 (0.08)*                      | 0.01 (0.01)*                |

Note: The table shows the cumulative percent response to a 25bp contractionary monetary shock from quarters 0 to 6. Estimates of Consumption Difference are the difference in area under the impulse response of married households ( $\Delta c_M$ ) and adjusted single households ( $\Delta c_S$ ) in Appendix Table H.1. LFP Spouse is the cumulative married labor force participation rate for the group of the same identified race and opposite sex (i.e., for the White Men group, LFP Spouse is the LFP response for married white women) in Figure 7c. Ratio  $\xi$  is calculated as LFP Spouse relative to Consumption Difference. Newey–West HAC standard errors in parentheses. \* Significant at one standard deviation. Data used for Consumption Difference is sourced from the CE and for LFP Spouse is sourced from the CPS.

The role of spousal labor supply is less apparent among households headed by women. Among households headed by white women, the married-single consumption difference is close to zero and male spouses reduce LFP. This suggests any consumption insurance for these households operates through margins other than spousal labor supply, such as asset holdings

<sup>20</sup>Appendix L reports cumulative responses at all horizons. The estimates are robust in the medium-term, as consumption and LFP differences are reduced in the long term.

or the composition of household earnings. Among households headed by black women, both the consumption difference and male-spouse LFP response are positive, although the latter is smaller than the response of black women spouses in households headed by men.

These patterns suggest that marriage alters the transmission of monetary policy shocks across demographic groups, but the mechanisms through which households insure against these shocks differ by household composition. The added earner margin can facilitate consumption smoothing, but does so by shifting the cost of the monetary shock from household resources to spouses' time allocation.<sup>21</sup> The weaker positive labor supply response of male spouses leaves households headed by women with less added earner support, suggesting that the stabilizing effects of marriage are mediated by gendered labor supply responses and differences in household access to other forms of insurance such as wealth.

## 6 Consumption-to-income response ratio

In this section, I calculate the household consumption-to-income response ratio to a contractionary monetary policy shock. This measure is defined as the ratio of the consumption impulse response to the income impulse response at each horizon following a 25bp contractionary shock. The responses of consumption are those for total, durable goods, or non-durable goods and services estimated in Figures 1 and 2, while the responses of income are those for total household income after tax for the overall sample.

This ratio captures the relative percentage response of consumption to the percentage response of income. Panel A of Table 4 shows that the ratio for total consumption is 1 in the first period following the shock, 1.17 after 6 quarters, and 0.21 after 12 quarters. The concentration of the consumption response in the short term is reflected in the large reduction in durable goods spending shown in Panel B and consumption reallocation towards non-durable goods and services shown in Panel C, which is reduced in the long term. The positive total consumption ratio is driven by the response of durable goods, which reaches 12.18 after 6 quarters, while that of non-durable goods and services is a modest -0.56.

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<sup>21</sup>Spouses may also bear additional unpaid domestic work in addition to employment. Women tend to bear the majority of the house and care work in married households, even in households where both spouses are employed (Thébaud 2010).

Converted to dollar terms, these ratios imply that for every dollar of income lost following a 25bp contractionary shock, households reduce overall spending by approximately 20 cents after 6 quarters. This adjustment consists of a 30-cent cut on durable goods and an 8-cent increase on non-durable goods and services.<sup>22</sup>

Table 4: Estimated consumption-to-income response ratios

| Horizon                               | All            | White<br>Men  | White<br>Women | Black<br>Men  | Black<br>Women    |
|---------------------------------------|----------------|---------------|----------------|---------------|-------------------|
| Panel A: Total consumption            |                |               |                |               |                   |
| 0                                     | 1.00 (0.96)*   | 0.66 (0.38)** | -1.22 (1.67)   | 0.61 (0.25)** | -11.86 (89.29)    |
| 6                                     | 1.17 (0.97)*   | 2.27 (4.83)   | 0.85 (0.93)    | -0.44 (0.35)* | 0.34 (0.18)**     |
| 12                                    | 0.21 (0.06)**  | 0.16 (0.05)** | 0.31 (0.10)**  | 0.32 (0.06)** | -0.04 (0.06)      |
| Panel B: Durable goods                |                |               |                |               |                   |
| 0                                     | 9.61 (9.01)*   | 5.32 (2.88)** | -16.48 (21.44) | -0.38 (0.63)  | -133.07 (1001.57) |
| 6                                     | 12.18 (10.07)* | 30.89 (63.49) | 12.16 (13.38)  | 7.33 (5.31)*  | 5.75 (2.50)**     |
| 12                                    | 1.91 (0.53)**  | 2.46 (0.65)** | 2.39 (0.59)**  | 1.24 (0.24)** | 0.64 (0.26)**     |
| Panel C: Non-durable goods & Services |                |               |                |               |                   |
| 0                                     | -0.23 (0.30)   | -0.05 (0.15)  | 0.52 (0.78)    | 0.73 (0.26)** | 1.53 (11.82)      |
| 6                                     | -0.56 (0.53)*  | -1.39 (2.98)  | -0.79 (0.86)   | -1.92 (1.35)* | 0.05 (0.10)       |
| 12                                    | 0.09 (0.04)**  | 0.06 (0.06)*  | 0.09 (0.03)**  | 0.14 (0.03)** | -0.02 (0.04)      |

Note: The table shows the ratio of the percent response of household consumption to total income to a 25bp contractionary monetary shock at various horizons. This ratio is computed for the consumption out of total goods, durable goods, and non-durable goods and services \*  $p < 0.32$ , \*\*  $p < 0.10$ . Data used is sourced from the CE.

Table 4 also presents the consumption-to-income response ratios for households disaggregated by gender and race. These ratios are directly comparable only within demographic groups, as each ratio is calculated relative to a group-specific change in income. Households headed by black women exhibit a significantly positive ratio after 6 quarters, driven primarily by a durable goods ratio of 5.75. Other households display clearer evidence of consumption reallocation after 6 quarters, but decrease consumption out of income losses in the longer run. For example, households headed by black men have a non-durable goods and services spending ratio of -1.92 after 6 quarters, indicating increased spending on these goods despite

<sup>22</sup>I convert the response ratio to dollar terms by multiplying the response ratio by the ratio of median household spending to median household income calculated from CE data in Table 1. The 20 cent cut of total goods spending is calculated as  $1.17 \times (4,136.9 / 24,686) = 0.196$ . The 30 cent cut of durables goods spending is calculated as  $12.18 \times (615 / 24,686) = 0.303$ , and the 8 cents increased on non-durable goods and services is calculated as  $-0.56 \times (3,521.8 / 24,686) = -0.079$ .

a decline in income, but display a significantly positive ratio for total consumption after 12 quarters.<sup>23</sup>

## 7 Robustness

I conduct several additional robustness checks to verify the results presented in previous sections. I first test the sensitivity of results to the monetary policy shock specification. I substitute the instrument in equation 1 for the Jarocinski and Karadi (2020) monetary policy shock series. I follow Jarocinski and Karadi (2020) in using the shock series to instrument the mean quarterly 1-year Treasury yield to ensure a strong instrument in the regression. As a second alternative specification, I instrument the Wu and Xia (2016) shadow rate series with the Bauer and Swanson (2023) monetary policy shock in equation 1. I aggregate the shadow rate series to a quarterly frequency by the sum of all shocks within the quarter. Both alternative specifications are estimated starting in 1990 due to data availability. Appendix M displays the response of gender and racial gaps in consumption, income, and unemployment for the baseline and two alternative specifications. The estimates are consistent in direction, magnitude, and statistical significance.

I then check whether the results are affected by different specifications of equation 1 such as OLS, lag lengths, and controls. I confirm that the main results using local projections-instrumental variables are consistent with local projections-OLS. The main results in Figure 2 are also similar up to four lags, as shown in Appendix Figure N.1. Including five or more lags of the controls and shocks substantially increases estimation uncertainty, resulting in wide and unstable confidence intervals at longer horizons as the effective sample size declines in the local projection regressions. Using less than five lags is consistent with the local projections literature. I next test for the importance of the vector of controls  $Z$  in equation 1 by running the local projections without this vector. The majority of results are consistent. However, it is important to include all  $Z$  controls in the estimation to separate out the effects of macroeconomic conditions. Omitting all or some of the controls likely

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<sup>23</sup>The negative total consumption ratio for households headed by black men reflects both consumption reallocation, where these households increase their spending on these goods even when their income decreases after a shock, and an increase in household income in the short run.

produces incorrect estimates.

Additionally, I confirm that the estimates are not sensitive to the method used to adjust the CE consumption series. Appendix Table N.1 shows that the estimates are similar when the consumption series are seasonally adjusted using the Census X-11 procedure without applying the four-quarter moving average. The resulting estimates have similar signs and magnitudes to the baseline estimates, although they are more volatile, particularly for households headed by black men and black women. These estimates indicate that the main findings do not arise from the smoothing imposed by the four-quarter moving average.

I also verify that the consumption results are similar for different periods. I separate the sample pre- and post-2001 and 2008 to study recessions and whether using shocks only from the unconventional monetary policy period affect the estimates. I find that the results for the full sample are qualitatively consistent with those using data pre-2001, pre-2008, or post-2008. Consumption of all goods and durable goods falls more for black than white households in the short run.

## **7.1 Monetary Policy Asymmetry**

I test for various forms of asymmetry in the transmission of monetary policy. There is much evidence in the literature (see Furceri et al. (2018), Jordà et al. (2020) and references cited therein) that the effects of monetary policy shocks are weaker in periods of low inflation and low growth, and that responses are weaker to expansionary versus contractionary shocks.

I first test the possible state dependence of the impulse response functions. I differentiate between periods of high and low inflation and economic booms and slumps. Next, I test the asymmetry in the direction of the monetary policy shock; positive (contractionary) versus negative (expansionary). Each series has 128 quarterly observations over the sample period. Of these quarters, 82 are classified as low inflation, 19 are slumps (recessions), and 78 are contractionary shock periods. I run local projections for each demographic group, including an indicator for each state of the economy. The below equation is estimated:

$$x_{g,t+h} = \alpha_{g,h} + \psi_{g,h}(L)Z_{t-1} + \beta_{g,h}Y_t^{2y}D_t + \beta_{g,h}Y_t^{2y}(1 - D_t) + \varepsilon_{g,t+h} \quad (2)$$

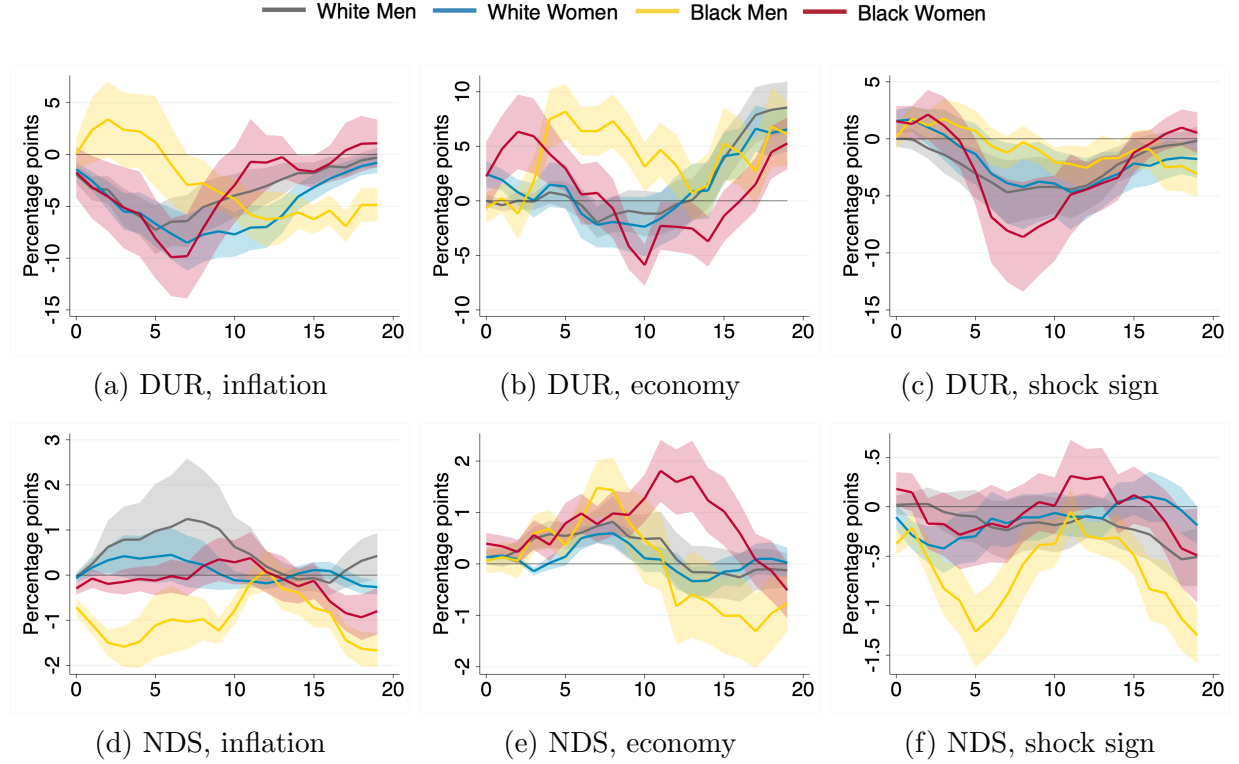
for  $h = 0, \dots, H - 1$ ,

where all variables are specified as in equation 1 and  $D$  is the state indicator. The interaction of the 2-year Treasury yield and  $D$  is instrumented with the interaction of the high-frequency monetary policy surprise measure and  $D$ . The first-stage results indicate that the instrument is sufficiently strong across all specifications except during low-inflation periods (see Appendix Table B.3). Accordingly, estimates for low-inflation periods should be interpreted with greater caution than those for other state comparisons. I assess the robustness of the results to alternative smooth-transition measures of economic and inflation states in Appendix O. Across the majority of specifications, the direction and magnitude of the estimated responses are consistent with those obtained using the state indicator specification.

The first test of state dependence is on inflationary periods. I follow Jordà et al. (2020) in constructing variable  $D$  as a dummy that equals one when the economy is in low inflation (at or below 2%) and zero in high inflation. I normalize the monetary policy shock in each period to be a 25bp rate hike. Figure 8a displays the difference in durable goods consumption responses to a contractionary shock between each inflation period. The difference is calculated as the absolute value of estimates for low inflation minus the absolute value of estimates for high inflation. This difference is negative for all households for durable goods in the short run, meaning that households decrease their durable goods consumption more in high than low inflation periods following a contractionary shock. The consumption responses are also significantly different between inflation periods for all households. Contractionary policy therefore has a larger negative impact on consumption during high inflation periods.

The second test of state dependence is on economic slumps and booms. I follow Jordà et al. (2020) in setting the indicator variable equal to one in an economic recession (slump) and zero otherwise. Also, in normalizing the shock in each state to be a 25bp contractionary shock. I calculate the difference between states as the absolute value of estimates for slumps minus the absolute value of estimates for booms. Figure 8b shows that durable goods consumption responses do not clearly vary between economic states for white households.

Figure 8: Difference in household consumption responses to monetary policy shocks by state and shock sign



Note: The figure shows differences in household consumption responses to monetary policy shocks, separately by households headed by white men, white women, black men, and black women. The first row is the response of durable goods (DUR) and the second row is of non-durable goods and services (NDS). Subfigures a and d show the effect of a 25bp contractionary shock on the absolute value consumption responses for low inflation ( $\leq 2\%$ ) minus high inflation ( $> 2\%$ ) and subfigures b and e for slumps (recessions) minus booms (non-recessions). Subfigures c and f show the difference in consumption responses to a 25bp contractionary (+) versus expansionary (-) monetary policy shock. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Household data used is sourced from the CE.

However, among households headed by black women the fall in durables consumption after a 25bp contractionary shock is larger in slumps by 6.3pp after 3 quarters, but in booms by 5.9pp after 10 quarters.

Lastly, I estimate whether monetary policy shocks have asymmetric effects on consumption by the direction of the shock. I set  $D$  to one if the shock is contractionary (positive) and to zero otherwise as in Furceri et al. (2018). For these estimates, the shocks are each scaled to be 25bp. Durables consumption falls in response to contractionary shocks and rises in response to expansionary shocks. Figure 8c shows that consumption falls more in

response to contractionary shocks than it rises in response to expansionary shocks. After 6 quarters, consumption falls by 6.8pp, 3.7pp, and 3.0pp more during contractionary than expansionary shocks for households headed by black women, white men and women, and black men respectively. The difference in consumption responses to contractionary versus expansionary monetary policy shocks is statistically significant for all households, see Appendix P. The estimates in section 3.1 are therefore driven by contractionary rather than expansionary shocks.

Consumption reallocation is suggested to occur more following contractionary shocks in high inflation periods, in booming economic periods, and in response to contractionary policy. Figures 8d-f display the difference in non-durable goods and services consumption responses to a contractionary shock in each state. Black households, especially those headed by black men, increase non-durables and services consumption more in high than low inflation periods in the short and medium term. All households increase their non-durables and services spending in economic booms, but in fact decrease it in economic slumps in the short term. This result implies that consumption reallocation is only a common behavior in economic booms, which could be driven by additional financial constraints faced by households during recessions.

Figure 8 also shows that the magnitude of the asymmetric transmission of shocks in each state differs by household group. The consumption responses of black households, either of durables for those headed by black women or non-durable goods and services for those headed by black men, are the largest in either state and to either shock sign, although with varying levels of statistical significance.

## 8 Conclusion

This paper is the first to estimate gender and racial gaps in household consumption responses to monetary policy shocks with this methodology and extended time period, as far as I know. It is also novel in explaining consumption responses through estimated impacts on labor market and income outcomes with spousal insurance theory. Consumption behaviors are influenced by total disposable income, which in turn is determined by access to education, employment, and financial opportunities. These other forms of inequality, which are more

often faced by people of color and women, are masked by measures of income inequality. Since income gaps obscure these other inequalities that affect how monetary policy is transmitted, focusing solely on income inequality has limited previous research.

I fill this gap in the literature by separately studying the consumption responses of households headed by white men, white women, black men, and black women while accounting for various household characteristics. The results show that falls in consumption following contractionary monetary policy shocks occur alongside rising unemployment rates and falling incomes. However, black households experience the largest negative effects of contractionary shocks in terms of durable goods expenditure and income reductions and higher unemployment. Neither household education, homeownership status, nor income explain gender and racial consumption gaps. However, analysis of household composition supports the theory of spousal insurance for consumption smoothing against shocks. Lastly, I find evidence that contractionary monetary policy shocks widen inequalities that are not reversed by expansionary shocks.

The findings in this paper are relevant for central bankers who might set interest rate policy to affect household spending. Policymakers must understand the determinants of aggregate outcomes. Also, how responses differ between positive and negative rate shocks to know whether monetary policy changes are having their intended effects.

More research is needed to understand how monetary policy affects consumption inequality by gender and race. This paper provides a holistic understanding of the interaction of these inequalities with conventional channels of monetary policy transmission. However, households headed by black women are still disproportionately impacted by policy shocks within each of these channels. A study could investigate the contribution of each channel with microeconomic data.

Additional efforts are also needed to incorporate gender and racial heterogeneity into models of monetary policy. My empirical estimates can inform model parameters to reflect heterogeneity in labor market outcomes, income, consumption, and MPCs. The results also provide motivation for quantitative models to explore nonlinearities in monetary policy shocks, especially when studying expansionary policy.

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Supplementary Material to:

“Monetary Policy Transmission to Consumption:  
Inequalities by Gender and Race”

By Aina Puig

## A Appendix: CE Data Cleaning

This section describes the data cleaning process for the Consumer Expenditure Survey (CE). The CE raw data files are available for download on the Bureau of Labor Statistics website. The expenditures data is aggregated from the MTAB files and the income data is constructed from the FMLI and MEMI files.

To create a quarterly expenditure series, expenditures are aggregated across months within the reported quarter. Observations are dropped if they report negative consumption on durable or non-durable goods and services. They are also dropped if they report negative net income data. I adjust for outliers by dropping households in the top or bottom 1% of expenditures in each quarter. This deletes about 3% of households. Since I am interested in studying how mortgage debt affects consumption, I also drop households that change their mortgage debt status during the year they are interviewed, following Cloyne et al. (2020). I only keep households whose head identifies as white or black race, given the small percentage of other racial groups in the sample. Following Cloyne et al. (2020), I also drop households whose heads are below 25 or over 75 years old, and keep households that were interviewed less than for their full survey period.

Categorization of consumption and income from the MTAB and MEMI CE files follows the classifications in Coibion et al. (2017). Expenditures on housing (including mortgage principle and interest, rental payments, and housing insurance), debt owed to creditors, taxes, sales of goods, rented goods, and subsidies are omitted from the sample. Data prior to 2004 are imputed according to Coibion et al. (2017) to account for CE imputations post-2004. Sample breaks and bracketed income values are also handled according to Coibion et al. (2017).

I adjust the quarterly expenditure series using a four-quarter moving average to reduce seasonal and high-frequency variation, which is particularly pronounced in the disaggregated demographic series. This specification preserves the medium-run movements relevant for estimating the dynamic response to monetary policy shocks. In Appendix Table N.1, I confirm consistent estimates when the consumption series are seasonally adjusted using the Census X-11 procedure without applying the four-quarter moving average.

For comparability over time, expenditure and income variables are deflated by the CPI to 2019 prices. To control for differences in household size, I adjust expenditures by the OECD scale of effective household size. All series are therefore expressed in real terms and adjusted for household size and composition. I additionally weight the series by the CE household weights.

The following tables display several average quarterly observation counts. Table A.1 contains three values representing average quarterly counts of each outcome of interest by household demographic group. The first value is the the average quarterly unweighted count, the second is the weighted share, and the third is the effective sample size by group over the sample date. Table A.2 shows that the majority of expenditure observations are positive expenditures, with all households reporting some expenditure in each quarter.

Table A.1: Average quarterly counts by household group

| Outcome    | Sample     | White<br>Men     | White<br>Women   | Black<br>Men   | Black<br>Women |
|------------|------------|------------------|------------------|----------------|----------------|
| Expend.    |            | 2289/0.52/1912   | 1628/0.35/1409   | 234/0.06/199   | 330/0.07/286   |
| Expend.    | No college | 977/0.23/811     | 693/0.15/594     | 128/0.03/108   | 175/0.04/150   |
| Expend.    | College    | 1312/0.29/1102   | 935/0.20/816     | 106/0.02/92    | 155/0.03/137   |
| Expend.    | Mortgagor  | 1159/0.26/965    | 698/0.15/608     | 87/0.02/74     | 94/0.02/82     |
| Expend.    | Owner      | 522/0.13/442     | 390/0.09/339     | 35/0.01/31     | 46/0.01/40     |
| Expend.    | Renter     | 588/0.13/493     | 529/0.11/454     | 110/0.03/94    | 187/0.04/161   |
| Expend.    | Income Q1  | 456/0.10/382     | 420/0.09/363     | 72/0.02/61     | 151/0.03/130   |
| Expend.    | Income Q2  | 384/0.09/325     | 328/0.07/282     | 48/0.01/41     | 74/0.02/64     |
| Expend.    | Income Q3  | 442/0.10/369     | 306/0.07/265     | 45/0.01/39     | 48/0.01/42     |
| Expend.    | Income Q4  | 482/0.11/402     | 294/0.06/256     | 38/0.01/33     | 35/0.01/31     |
| Expend.    | Income Q5  | 525/0.12/438     | 280/0.06/244     | 31/0.01/27     | 23/0.00/20     |
| Expend.    | Single     | 660/0.15/560     | 912/0.20/780     | 109/0.02/94    | 261/0.06/226   |
| Expend.    | Married    | 1629/0.37/1353   | 716/0.15/630     | 126/0.03/106   | 69/0.02/60     |
| Income     | Single     | 855/0.16/721     | 1264/0.24/1074   | 130/0.03/110   | 313/0.06/270   |
| Income     | Married    | 1798/0.34/1486   | 711/0.13/622     | 133/0.03/111   | 66/0.01/58     |
| - Salary   | Single     | 931/0.16/788     | 1365/0.23/1164   | 141/0.02/120   | 337/0.06/290   |
| - Salary   | Married    | 1944/0.35/1614   | 829/0.13/727     | 144/0.03/121   | 77/0.01/67     |
| - Business | Single     | 909/0.16/770     | 1345/0.24/1146   | 138/0.02/118   | 333/0.06/288   |
| - Business | Married    | 1880/0.34/1561   | 792/0.13/695     | 140/0.03/118   | 75/0.01/65     |
| - Finance  | Single     | 911/0.16/771     | 1335/0.24/1137   | 139/0.02/118   | 333/0.06/287   |
| - Finance  | Married    | 1897/0.34/1574   | 796/0.13/698     | 141/0.03/118   | 75/0.01/65     |
| Savings    | Single     | 877/0.16/742     | 1274/0.24/1086   | 133/0.02/113   | 318/0.06/275   |
| Savings    | Married    | 1801/0.34/1494   | 762/0.13/669     | 133/0.03/111   | 70/0.01/61     |
| Unemp      |            | 29225/0.47/22744 | 25501/0.40/19822 | 2770/0.06/2215 | 3337/0.07/2716 |
| LFP        | Single     | 16923/0.18/13166 | 20062/0.21/15746 | 2640/0.04/2090 | 4044/0.05/3259 |
| LFP        | Married    | 24005/0.24/18773 | 24179/0.24/18968 | 1751/0.02/1409 | 1721/0.02/1408 |

Note: The table shows the average quarterly unweighted count, weighted share, and effective sample size by group over the sample dates. First rows are expenditures from CE data, which are used to compute estimates for total, durable goods, and nondurable goods and services in Figures 2-5. The next rows are income from CE data, used in Figure 6, and the unemployment rate and labor force participation rate from CPS data, used in Figure 7.

Table A.2: Average quarterly counts of households with positive expenditures

|                           | White<br>Men | White<br>Women | Black<br>Men | Black<br>Women |
|---------------------------|--------------|----------------|--------------|----------------|
| Durables                  | 1342/0.58    | 899/0.52       | 106/0.46     | 137/0.41       |
| Non-durables & Services   | 2289/1.00    | 1628/1.00      | 234/1.00     | 330/1.00       |
| Non-durables              | 2289/1.00    | 1628/1.00      | 234/1.00     | 330/1.00       |
| - Clothing, footwear      | 1670/0.72    | 1134/0.72      | 169/0.72     | 226/0.71       |
| - Food                    | 2288/1.00    | 1627/1.00      | 234/1.00     | 329/1.00       |
| - House furnishings       | 984/0.42     | 664/0.41       | 64/0.27      | 85/0.26        |
| - Tech., entert.          | 2265/0.99    | 1608/0.99      | 229/0.98     | 321/0.97       |
| - Transportation          | 2186/0.95    | 1493/0.91      | 199/0.85     | 243/0.72       |
| Services                  | 2281/1.00    | 1621/1.00      | 232/0.99     | 326/0.99       |
| - Care                    | 1791/0.77    | 1136/0.70      | 153/0.66     | 186/0.57       |
| - Utilities, maintenance  | 2178/0.95    | 1543/0.95      | 214/0.91     | 299/0.90       |
| - Insurance               | 1980/0.86    | 1381/0.84      | 183/0.78     | 240/0.72       |
| - Tech., ed., and entert. | 1959/0.85    | 1357/0.81      | 168/0.72     | 225/0.67       |
| - Transportation          | 1646/0.71    | 1089/0.67      | 154/0.66     | 200/0.62       |

Note: The table shows average quarterly cell counts and within-group weighted share of positive expenditures by group over the sample dates. Data used is sourced from the CE.

## B Appendix: LP-IV first-stage F-statistics

This section presents first-stage F-statistics of estimates using the local projections instrumental variables specifications in equations 1 and 2. Each cell within the tables below is a separate regression listed in the order of figures in the main text. F-statistics of estimates of aggregate outcomes in Figure 1 are presented in Table B.1, of main outcomes in Table B.2, and of additional outcomes in Table B.3.

Table B.1: First-stage F-statistics of aggregate outcomes

| Outcome                 | Statistic |
|-------------------------|-----------|
| Industrial production   | 16.81     |
| Unemployment rate       | 32.98     |
| Total income            | 14.59     |
| Total consumption       | 15.89     |
| Consumption Gap (NDS-D) | 15.16     |

Note: Estimates of aggregate variables presented in Figure 1, see text.

Table B.2: First-stage F-statistics of main outcomes by group

| Outcome                   | Sample     | White<br>Men | White<br>Women | Black<br>Men | Black<br>Women |
|---------------------------|------------|--------------|----------------|--------------|----------------|
| Total consumption         |            | 16.29        | 18.18          | 22.99        | 19.07          |
| Durables                  |            | 16.42        | 12.85          | 14.56        | 16.49          |
| Non-durables & Services   |            | 21.69        | 26.61          | 34.44        | 14.51          |
| Durables                  | No college | 15.41        | 15.09          | 15.30        | 18.83          |
| Durables                  | College    | 17.17        | 13.13          | 14.95        | 15.33          |
| Durables                  | Mortgagor  | 14.50        | 13.61          | 16.28        | 16.44          |
| Durables                  | Owner      | 16.16        | 15.03          | 13.84        | 16.93          |
| Durables                  | Renter     | 21.84        | 14.65          | 13.82        | 17.57          |
| Durables                  | Income Q1  | 16.72        | 15.08          | 14.53        | 17.75          |
| Durables                  | Income Q2  | 14.13        | 13.66          | 15.07        | 17.43          |
| Durables                  | Income Q3  | 18.81        | 18.92          | 18.81        | 18.17          |
| Durables                  | Income Q4  | 16.49        | 19.05          | 16.63        | 17.40          |
| Durables                  | Income Q5  | 17.05        | 14.92          | 16.77        | 16.83          |
| Durables                  | Single     | 15.39        | 14.35          | 14.83        | 16.61          |
| Durables                  | Married    | 15.40        | 15.28          | 15.40        | 14.45          |
| Non-durables              |            | 22.43        | 36.40          | 43.67        | 16.78          |
| - Clothing, footwear      |            | 20.28        | 21.75          | 21.63        | 17.80          |
| - Food                    |            | 17.88        | 40.43          | 29.44        | 18.71          |
| - House furnishings       |            | 16.41        | 19.90          | 23.83        | 23.34          |
| - Tech., entert.          |            | 22.71        | 24.23          | 27.71        | 17.08          |
| - Transportation          |            | 14.79        | 15.17          | 19.41        | 13.42          |
| Services                  |            | 25.05        | 16.67          | 18.89        | 23.94          |
| - Care                    |            | 15.30        | 15.59          | 18.16        | 14.97          |
| - Utilities, maintenance  |            | 13.36        | 13.19          | 21.36        | 14.15          |
| - Insurance               |            | 17.94        | 18.82          | 17.79        | 20.47          |
| - Tech., ed., and entert. |            | 19.47        | 17.26          | 16.84        | 19.79          |
| - Transportation          |            | 20.80        | 16.63          | 13.36        | 16.49          |
| Non-durables & Services   | No college | 18.34        | 25.63          | 22.49        | 19.28          |
| Non-durables & Services   | College    | 19.90        | 19.41          | 35.46        | 16.71          |
| Non-durables & Services   | Mortgagor  | 16.11        | 19.38          | 25.46        | 16.55          |
| Non-durables & Services   | Owner      | 17.57        | 15.66          | 16.45        | 16.74          |
| Non-durables & Services   | Renter     | 25.17        | 23.99          | 23.80        | 19.28          |
| Non-durables & Services   | Income Q1  | 19.07        | 36.63          | 17.97        | 21.37          |
| Non-durables & Services   | Income Q2  | 25.98        | 33.03          | 19.06        | 28.11          |
| Non-durables & Services   | Income Q3  | 21.31        | 15.79          | 15.82        | 22.00          |
| Non-durables & Services   | Income Q4  | 16.16        | 25.62          | 18.54        | 18.06          |
| Non-durables & Services   | Income Q5  | 14.69        | 15.11          | 26.39        | 18.95          |
| Non-durables & Services   | Single     | 16.81        | 13.84          | 24.47        | 17.14          |
| Non-durables & Services   | Married    | 18.92        | 35.58          | 21.14        | 28.50          |

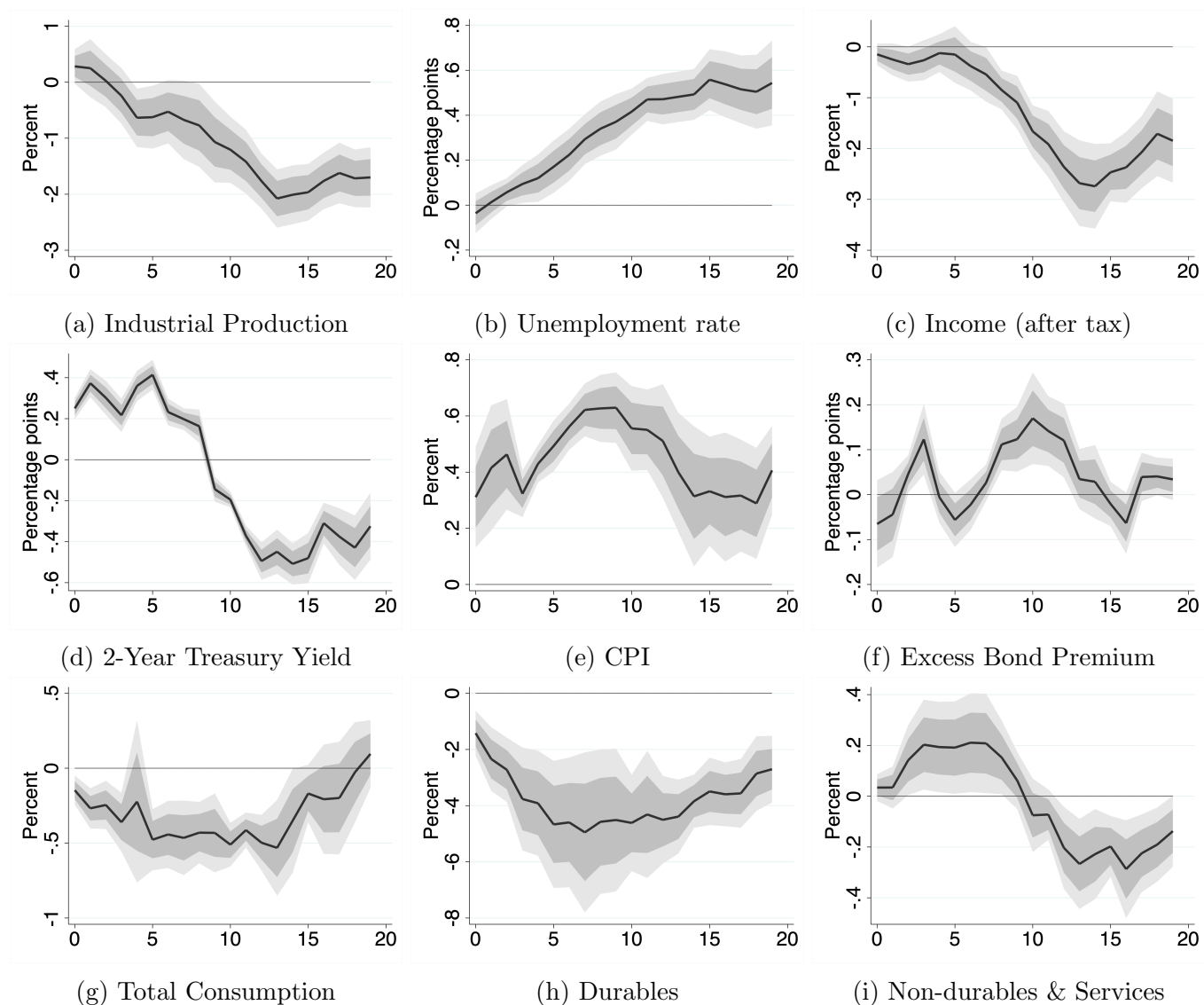
Note: See details in text.

Table B.3: First-stage F-statistics of additional outcomes by group

| Outcome                 | Sample               | White<br>Men | White<br>Women | Black<br>Men | Black<br>Women |
|-------------------------|----------------------|--------------|----------------|--------------|----------------|
| Total income            | Single               | 15.53        | 15.05          | 16.29        | 19.22          |
| Total income            | Married              | 14.79        | 15.10          | 15.58        | 17.33          |
| - Salary                | Single               | 17.54        | 15.07          | 16.14        | 16.26          |
| - Salary                | Married              | 14.35        | 18.16          | 17.17        | 20.76          |
| - Business              | Single               | 17.01        | 15.76          | 18.05        | 24.72          |
| - Business              | Married              | 13.67        | 14.12          | 15.64        | 15.20          |
| - Finance               | Single               | 14.74        | 15.72          | 20.53        | 24.35          |
| - Finance               | Married              | 15.56        | 14.88          | 16.21        | 12.48          |
| Savings                 | Single               | 19.86        | 13.66          | 15.38        | 21.46          |
| Savings                 | Married              | 11.77        | 19.64          | 16.22        | 18.09          |
| Unemployment rate       |                      | 30.77        | 30.30          | 33.92        | 28.13          |
| LFP rate                | Single               | 22.87        | 21.85          | 23.64        | 23.93          |
| LFP rate                | Married              | 27.27        | 24.88          | 26.84        | 26.61          |
| Durables                | Low inflation        | 2.24         | 3.17           | 1.65         | 1.65           |
| Durables                | High inflation       | 53.45        | 60.84          | 39.81        | 69.53          |
| Durables                | Economic slump       | 7.79         | 8.40           | 9.77         | 8.54           |
| Durables                | Economic boom        | 16.64        | 15.08          | 17.76        | 18.65          |
| Durables                | Contractionary shock | 150.81       | 138.38         | 96.90        | 118.32         |
| Durables                | Expansionary shock   | 49.15        | 74.60          | 71.94        | 68.73          |
| Non-durables & Services | Low inflation        | 1.09         | 1.63           | 2.53         | 1.80           |
| Non-durables & Services | High inflation       | 62.12        | 29.89          | 38.90        | 25.14          |
| Non-durables & Services | Economic slump       | 12.01        | 10.08          | 6.35         | 11.10          |
| Non-durables & Services | Economic boom        | 14.84        | 15.05          | 18.68        | 13.79          |
| Non-durables & Services | Contractionary shock | 143.59       | 197.24         | 360.74       | 146.97         |
| Non-durables & Services | Expansionary shock   | 105.23       | 99.80          | 103.81       | 90.04          |

Note: See details in text.

## C Appendix: Effects of monetary policy shocks on macroeconomic aggregates

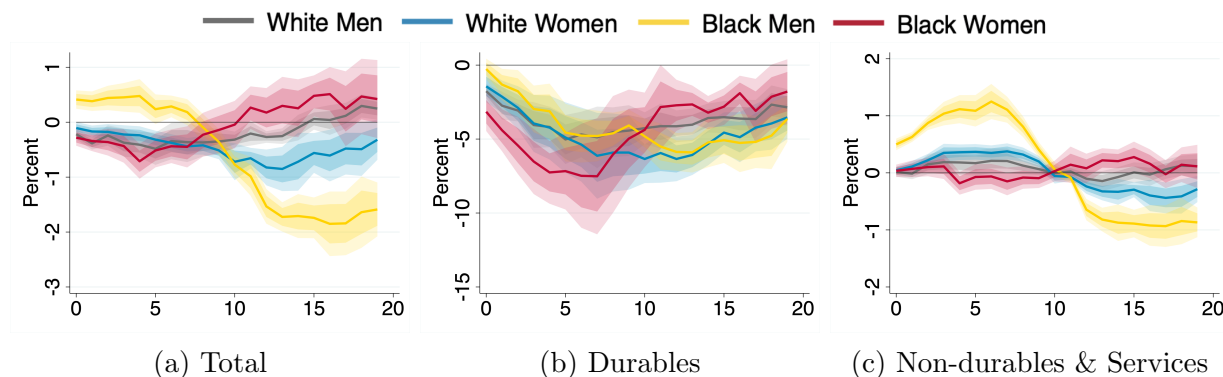


Note: The figure shows the effects of a 25bp contractionary monetary policy shock on aggregate variables. Time (horizontal axis) is in quarters. Dark and light grey shaded areas represent one standard deviation and 90% confidence intervals respectively. Data in subfigure (a) and (d) are sourced from the Board of Governors of the Federal Reserve System, (b) from the CPS, (c) and (g)-(i) from the CE, (e) from the BLS, and (f) from Favara et al. (2016).

## D Appendix: Differences in effects of monetary policy shocks on consumption

This section provides a formal assessment of differences in demographic group differences in household consumption impulse responses based on the results reported in Figure 2, displayed in Figure D.1 including 90% confidence interval bands.

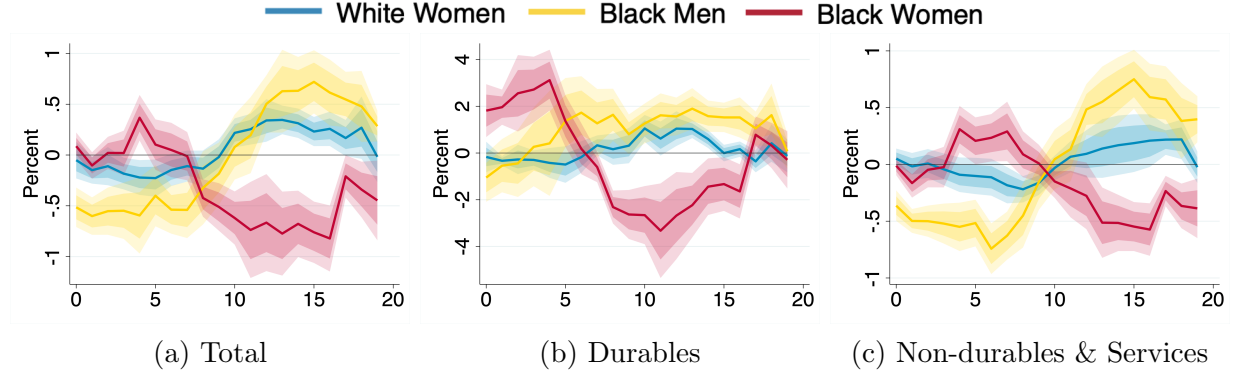
Figure D.1: Effects of monetary policy shocks on household consumption



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption, separately by households headed by white men, white women, black men, and black women. Sub-figures show consumption of (a) total goods, (b) durable goods, and (c) non-durable goods and services. Time (horizontal axis) is in quarters. Dark and light shaded areas represent one standard deviation and 90% confidence intervals respectively. Data used is sourced from the CE.

Below are two assessments of differences in responses. First, Figure D.2 shows the impulse responses of the gap in log consumption between each demographic group and households headed by white men to a 25bp monetary policy shock. Second, Table D.1 shows tests of the differences in responses of each demographic group and households headed by white men in Figure 2. Following Jordà et al. (2020), I first calculate the area under the impulse response of each group. Three areas are calculated, from quarter 0-6, 0-12, and 0-19, for comparison over distinct time trajectories. Then, I calculate the difference across demographic groups, reporting the estimated numerical value along with its standard error.

Figure D.2: Effects of monetary policy shocks on gap with households headed by white men



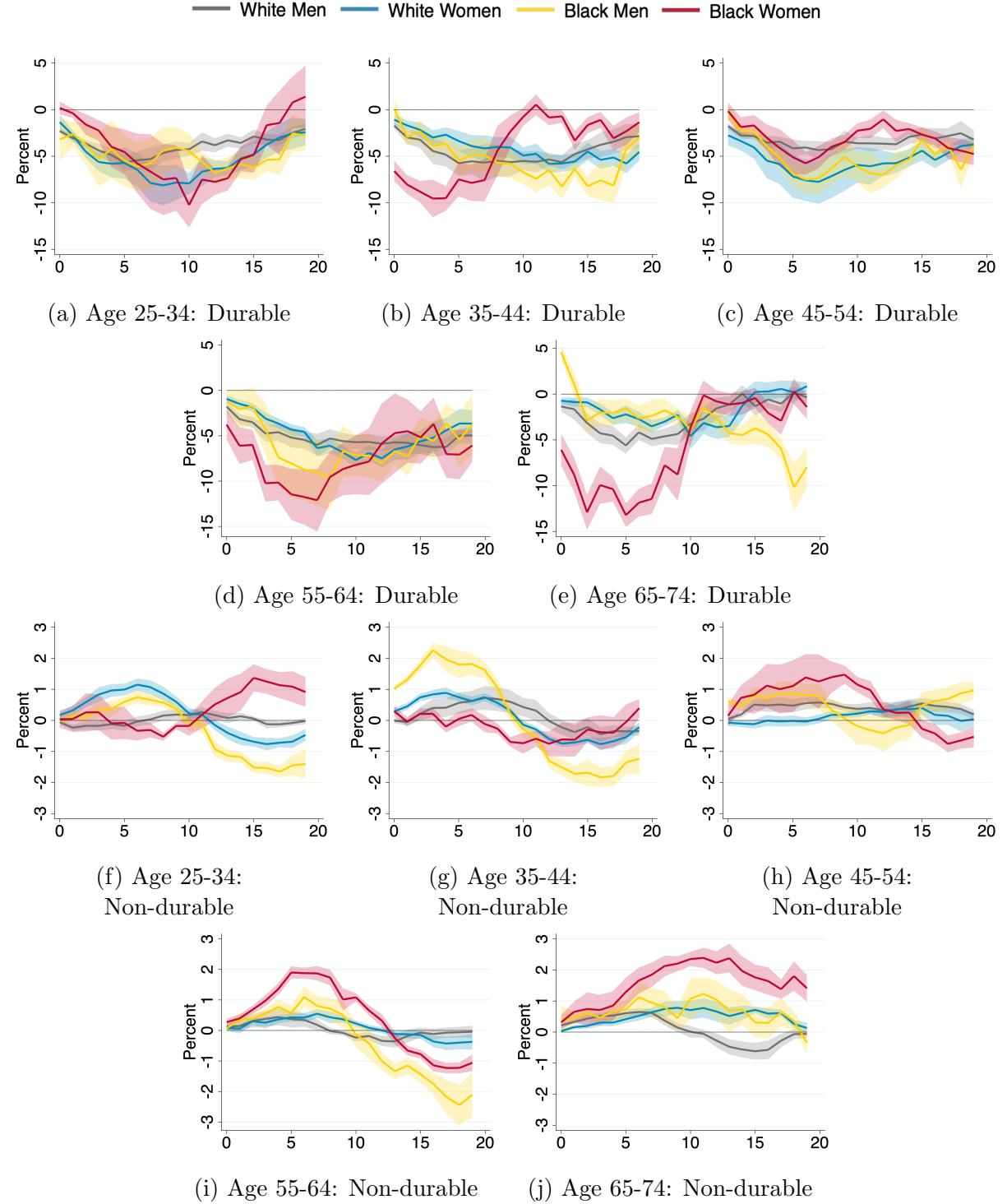
Note: The figure shows the effects of a 25bp contractionary monetary policy shock on gaps in consumption between households headed by white men and white women (blue), black men (yellow), and black women (red). Sub-figures show consumption of (a) total goods, (b) durable goods, and (c) non-durable goods and services. Time (horizontal axis) is in quarters. Dark and light shaded areas represent one standard deviation and 90% confidence intervals respectively. Data used is sourced from the CE.

Table D.1: Difference in responses of group and households headed by white men

| Response                | Group       | Difference summed over quarters |                |                |
|-------------------------|-------------|---------------------------------|----------------|----------------|
|                         |             | Qtr 0-6                         | Qtr 0-12       | Qtr 0-19       |
| Total                   | White Women | 1.25 (1.35)                     | -0.07 (2.20)   | -3.93 (2.63)*  |
|                         | Black Men   | 6.40 (1.56)*                    | 7.29 (2.71)*   | -1.82 (3.89)   |
|                         | Black Women | -1.94 (1.65)*                   | -3.46 (2.53)*  | -3.30 (3.11)*  |
| Durables                | White Women | 2.73 (8.30)                     | -5.47 (14.26)  | -14.04 (18.08) |
|                         | Black Men   | 15.84 (9.01)*                   | 16.23 (15.47)* | 4.87 (19.68)   |
|                         | Black Women | -12.38 (11.11)*                 | -11.78 (16.76) | -6.95 (19.51)  |
| Non-durables & Services | White Women | 0.61 (0.63)                     | 0.50 (1.07)    | -2.29 (1.52)*  |
|                         | Black Men   | 4.75 (1.30)*                    | 6.79 (1.80)*   | 1.61 (2.54)    |
|                         | Black Women | -1.89 (0.83)*                   | -4.90 (1.36)*  | -5.45 (1.93)*  |

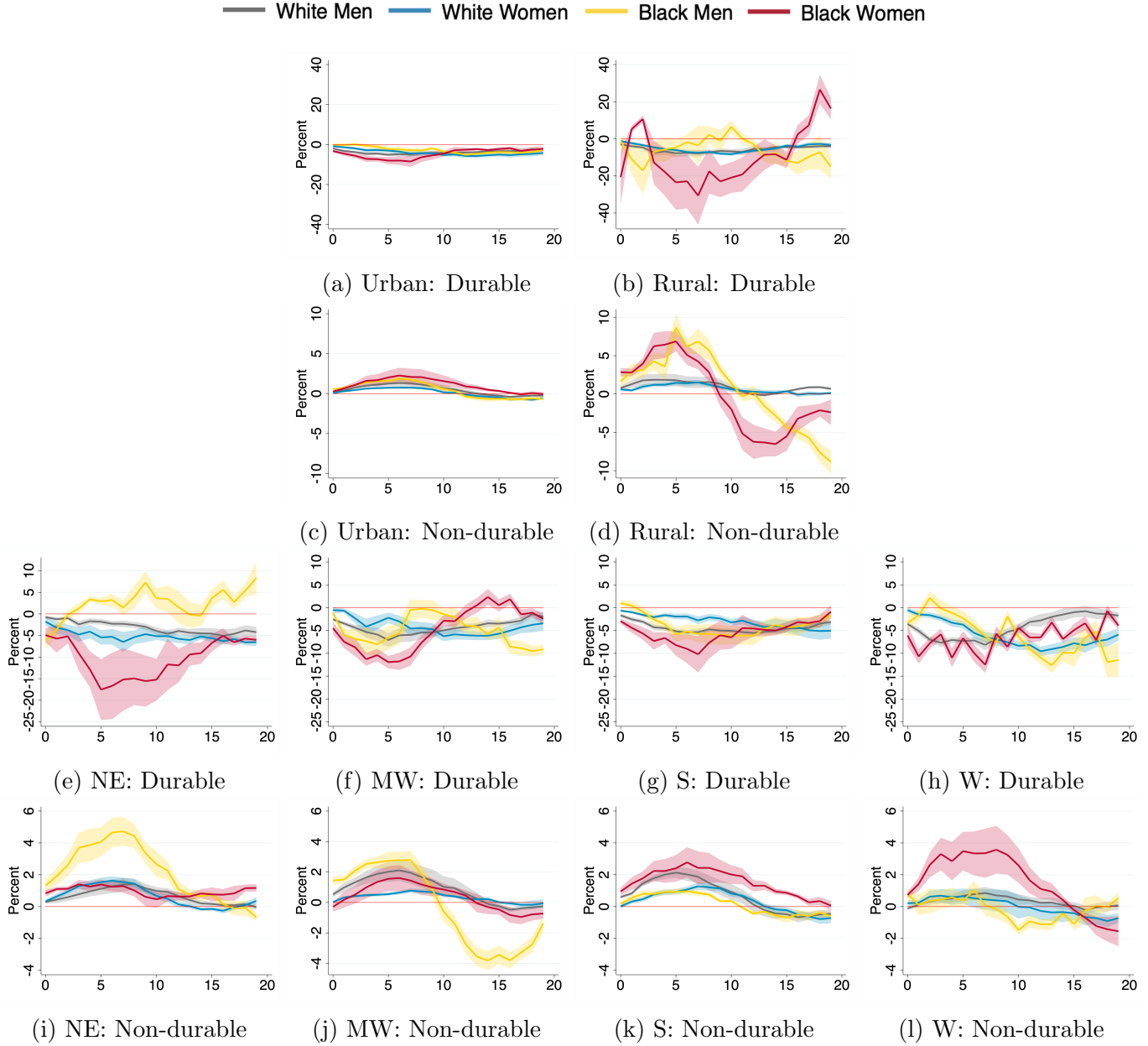
Note: The table shows the difference in area under the impulse response of each demographic group and households headed by white men in Figure 2. Differences are calculated at three time trajectories; from quarter 0-6, 0-12, and 0-19. \* Significant at one standard deviation. Newey–West HAC standard errors in parentheses. Data used is sourced from the CE.

## E Appendix: Effects of monetary policy shocks on household consumption by age



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption by the age of the head of household. The first two rows show responses of consumption of durable goods while the last two rows of non-durable goods and services. Each figure corresponds to households within a different age band. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

## F Appendix: Effects of monetary policy shocks on household consumption by geography



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption by geographic location. The first and third rows show responses of consumption of durable goods while the second and fourth rows of non-durable goods and services. The first two rows split the sample between urban and rural households. Sub-figures e-l show estimates for households located in the North-East (NE), Mid-West (MW), South (S), and West (W) of the U.S. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

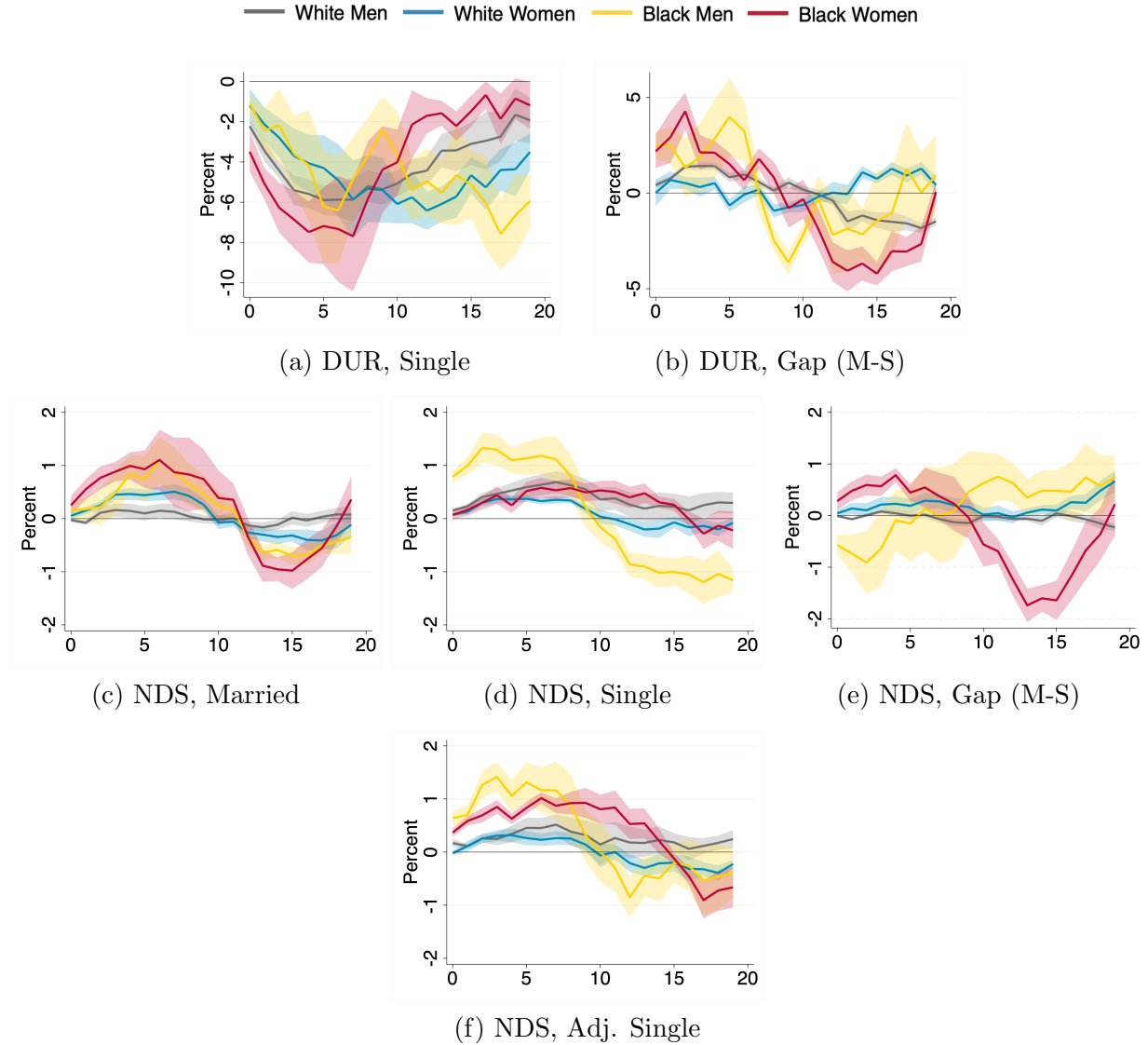
**G Appendix: Effect of monetary policy shocks on categories of household non-durable goods and services consumption, quarter 3**

|                           | White<br>Men    | White<br>Women  | Black<br>Men    | Black<br>Women  |
|---------------------------|-----------------|-----------------|-----------------|-----------------|
| Non-durables & Services   | 0.19 ( 0.12)*   | 0.35 ( 0.10)**  | 1.04 ( 0.13)**  | 0.11 ( 0.13)    |
| Non-durables              | 0.24 ( 0.10)**  | 0.49 ( 0.15)**  | 1.32 ( 0.19)**  | 0.40 ( 0.11)**  |
| - Clothing, footwear      | 0.45 ( 0.27)*   | 0.96 ( 0.22)**  | 0.91 ( 0.56)*   | -0.01 ( 0.53)   |
| - Food                    | 0.25 ( 0.10)**  | 0.45 ( 0.13)**  | 1.25 ( 0.24)**  | 0.69 ( 0.17)**  |
| - House furnishings       | -0.89 ( 0.35)** | 0.91 ( 0.37)**  | 1.52 ( 1.64)    | -3.00 ( 0.85)** |
| - Tech., entert.          | -0.33 ( 0.06)** | -0.37 ( 0.12)** | 0.67 ( 0.43)*   | -0.42 ( 0.18)** |
| - Transportation          | 0.38 ( 0.24)*   | 0.07 ( 0.15)    | 2.13 ( 0.35)**  | 0.38 ( 0.30)*   |
| Services                  | 0.41 ( 0.16)**  | 0.20 ( 0.14)*   | 0.77 ( 0.10)**  | -0.51 ( 0.22)** |
| - Care                    | 1.03 ( 0.35)**  | 0.91 ( 0.11)**  | 1.50 ( 0.32)**  | -1.15 ( 0.29)** |
| - Utilities, maintenance  | 0.88 ( 0.17)**  | 0.54 ( 0.20)**  | 1.96 ( 0.32)**  | 0.61 ( 0.17)**  |
| - Insurance               | -0.03 ( 0.10)   | -0.28 ( 0.31)   | 1.05 ( 0.74)*   | -1.23 ( 0.26)** |
| - Tech., ed., and entert. | 1.02 ( 0.38)**  | 1.45 ( 0.33)**  | 1.38 ( 0.45)**  | -1.03 ( 0.33)** |
| - Transportation          | -1.48 ( 0.35)** | -0.98 ( 0.20)** | -1.02 ( 0.33)** | -1.20 ( 0.24)** |

Note: The table shows the percent response of each household expenditure category to a 25bp contractionary monetary shock at quarter 3. Category labels are shortened for presentation: the fifth row of house furnishings includes other house non-durable goods; the sixth row is technology and entertainment non-durable goods; the ninth row includes child, elder, house, and personal care services; the tenth row is house utilities and maintenance services; and the twelfth row is technology, education, and entertainment services. Newey–West HAC standard errors in parentheses. \*  $p < 0.32$ , \*\*  $p < 0.10$ . Data used is sourced from the CE.

## H Appendix: Effects of monetary policy shocks on household consumption by marital status

Figure H.1: Effects of monetary policy shocks on household consumption by adjustment in marital status



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption of durable goods (DUR) and nondurable goods and services (NDS), separately by households headed by white men, white women, black men, and black women. The responses of unadjusted single households are in figures a and d, those of the gap between married (M) and unadjusted single (S) households are in figures b and e, those of married households in figure c, and adjusted single households in figure f. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

Table H.1: Difference in married and single household responses by group

| Response                | Group       | Difference summed over quarters |                 |                |
|-------------------------|-------------|---------------------------------|-----------------|----------------|
|                         |             | Qtr 0-6                         | Qtr 0-12        | Qtr 0-19       |
| Durables                | White men   | -12.28 (8.56)*                  | -18.31 (14.43)* | -2.57 (17.96)  |
|                         | White Women | -0.65 (9.78)                    | 5.66 (16.16)    | 6.26 (20.04)   |
|                         | Black Men   | -16.15 (10.84)*                 | -5.40 (17.97)   | 32.28 (23.19)* |
|                         | Black Women | -20.53 (14.14)*                 | -23.96 (20.47)* | -7.02 (23.48)  |
| Non-durables & Services | White men   | 0.68 (0.93)                     | 0.91 (1.54)     | 2.23 (1.95)*   |
|                         | White Women | -2.87 (0.93)*                   | -4.67 (1.31)*   | -6.07 (1.66)*  |
|                         | Black Men   | 2.34 (2.61)                     | 0.11 (3.81)     | -0.64 (4.94)   |
|                         | Black Women | -0.97 (1.88)                    | 5.86 (2.96)*    | 7.94 (3.88)*   |

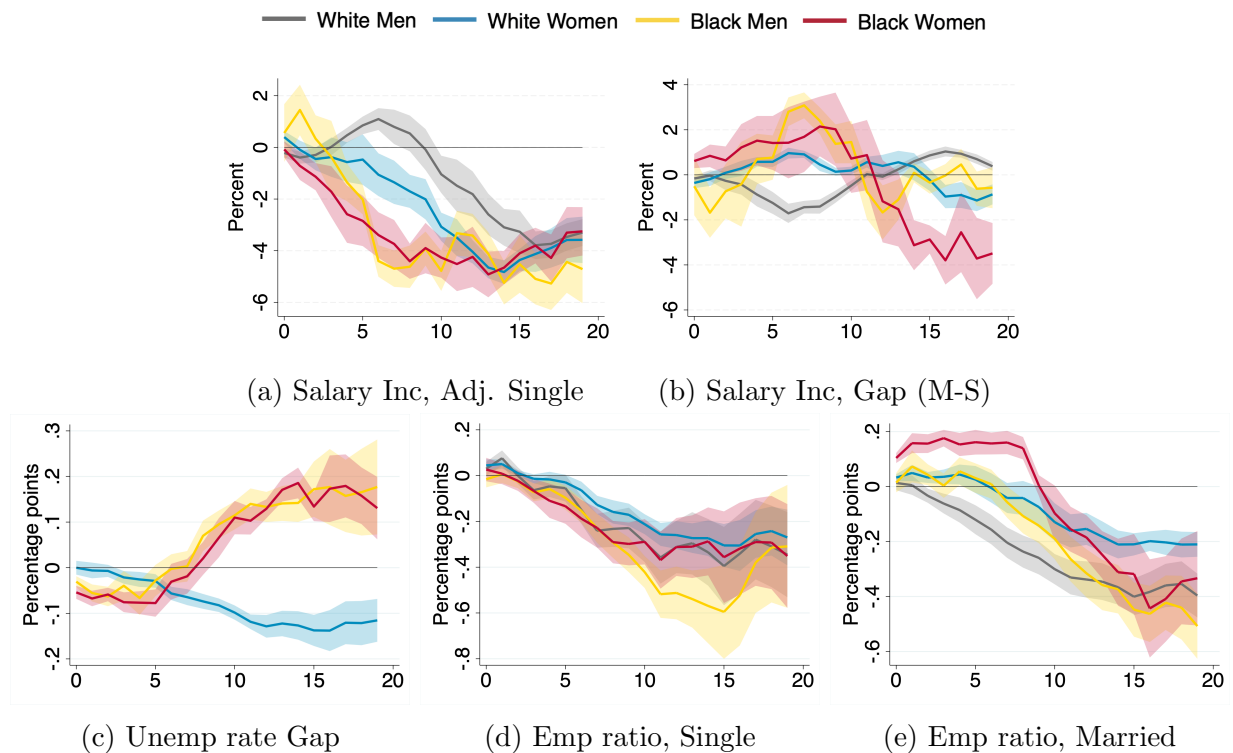
Note: The table shows the difference in area under the impulse response of each married and adjusted single household in Figure 5. Differences are calculated at three time trajectories; from quarter 0-6, 0-12, and 0-19. \* Significant at one standard deviation. Standard error in parentheses. Data used is sourced from the CE.

Table H.2: Difference in durables responses of group and households headed by white men by marital status

| Sample  | Group       | Difference summed over quarters |                 |                 |
|---------|-------------|---------------------------------|-----------------|-----------------|
|         |             | Qtr 0-6                         | Qtr 0-12        | Qtr 0-19        |
| Married | White Women | -4.46 (9.03)                    | -19.02 (15.59)* | -20.23 (19.05)* |
|         | Black Men   | 22.15 (9.09)*                   | 22.79 (15.72)*  | 15.07 (19.41)   |
|         | Black Women | -0.17 (10.48)                   | -1.80 (17.73)   | -9.70 (22.22)   |
| Single  | White Women | 9.09 (9.17)                     | 10.42 (15.16)   | -4.29 (19.07)   |
|         | Black Men   | 12.75 (10.77)*                  | 19.86 (17.01)*  | 24.59 (22.35)*  |
|         | Black Women | -6.96 (12.14)                   | -2.24 (17.63)   | -8.09 (20.37)   |

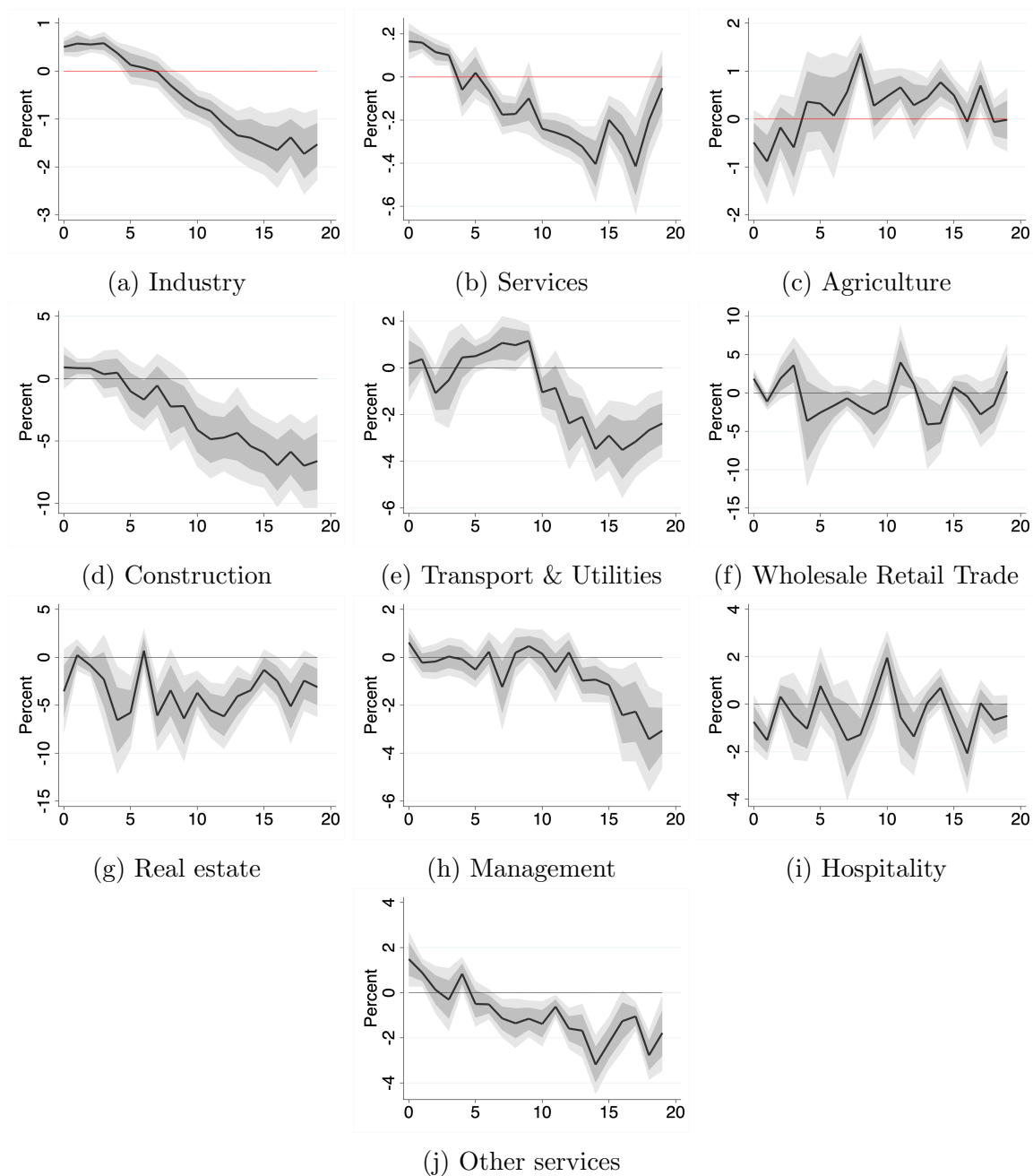
Note: The table shows the difference in area under the impulse response of each demographic group and households headed by white men for durable consumption by marital status in Figure 5. Differences are calculated at three time trajectories; from quarter 0-6, 0-12, and 0-19. \* Significant at one standard deviation. Standard error in parentheses. Data used is sourced from the CE.

# I Appendix: Effects of monetary policy shocks on additional labor market outcomes



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on labor market outcomes. The first row shows the effects on household salary income conditional on households that have an employed spouse from CE data. Figure a is for single households, adjusted with weights to match married household characteristics, and figure b is for the gap between married and single households. The second row shows the effects on individual labor market outcomes from CPS data. Figure c displays the gap between demographic groups and white men's unemployment rate. Figures d-e show the effects on the single and married individual employment-population ratio. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals.

## J Appendix: Effects of monetary policy shocks on employment by sector



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on log employment by sector. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data in figures a-c is from the ILO and d-j from the BLS 2000-2019.

Table J.1: Effects of monetary policy shocks on gaps in employment by sector

| Sector                           | Gap           |             |
|----------------------------------|---------------|-------------|
|                                  | White - Black | Men - Women |
| Construction                     | 8.218*        | -0.099      |
|                                  | (5.402)       | (1.969)     |
| Education & Health services      | 0.355         | 0.802*      |
|                                  | (1.276)       | (0.782)     |
| Entertainment                    | -2.764        | -0.186      |
|                                  | (6.21)        | (0.769)     |
| Finance & Insurance              | 0.536         | -0.558*     |
|                                  | (1.41)        | (0.528)     |
| Hospitality                      | 3.793*        | -1.172*     |
|                                  | (1.377)       | (0.436)     |
| Information                      | -3.708*       | 0.699       |
|                                  | (1.792)       | (0.796)     |
| Management                       | 3.76          | 1.062*      |
|                                  | (3.909)       | (0.817)     |
| Manufacturing                    | 2.661*        | -0.24       |
|                                  | (0.991)       | (0.512)     |
| Mining                           | 18.022*       | -0.542      |
|                                  | (5.654)       | (4.706)     |
| Other services                   | -0.498        | -1.091      |
|                                  | (1.122)       | (1.339)     |
| Professional & Business services | 1.694*        | -1.365*     |
|                                  | (1.292)       | (0.889)     |
| Public Administration            | -0.567        | 1.31*       |
|                                  | (0.955)       | (0.856)     |
| Real estate                      | 8.037*        | 0.237       |
|                                  | (1.767)       | (1.247)     |
| Transport & Utilities            | -1.689*       | 0.35        |
|                                  | (0.932)       | (2.026)     |
| Wholesale Retail Trade           | -0.501        | -1.2*       |
|                                  | (0.901)       | (1.089)     |

Note: The table shows the effects of a 25bp contractionary monetary policy shock on race and gender gaps in log employment by sector at quarter 4. \* Significant at one standard deviation. Standard deviation in parentheses. Data is from the BLS 2000-2019.

## K Appendix: Collective household model with endogenous labor supply

This section outlines a simple two-period collective household model for illustrative purposes. A household  $h$  consisting of spouses  $j \in \{1, 2\}$  chooses consumption  $c$  and the labor supply of its spouses  $h_j$  to maximize joint lifetime utility:

$$\begin{aligned} \max_{c_t, h_{1,t}, h_{2,t}} \quad & \mathbb{E}_0 \sum_{t=1}^2 \beta^{t-1} \left[ \frac{c_t^{1-1/\sigma}}{1-1/\sigma} - \sum_{j=1}^2 \psi_j \frac{h_{j,t}^{1+1/\phi_j}}{1+1/\phi_j} \right] \\ \text{subject to} \quad & c_1 + \frac{c_2}{1+r(\epsilon)} = A_1 + \sum_{j=1}^2 \left( w_j(\epsilon) h_{j,1} + \frac{w_j h_{j,2}}{1+r(\epsilon)} \right), \end{aligned}$$

where  $\sigma$  is the intertemporal elasticity of substitution and  $\phi_j$  is the Frisch elasticity of labor supply for spouse  $j$ .

The intratemporal optimality condition for the secondary earner ( $j = 2$ ) is given by:

$$h_{2,t}^{1/\phi_2} = \lambda_t w_{2,t}$$

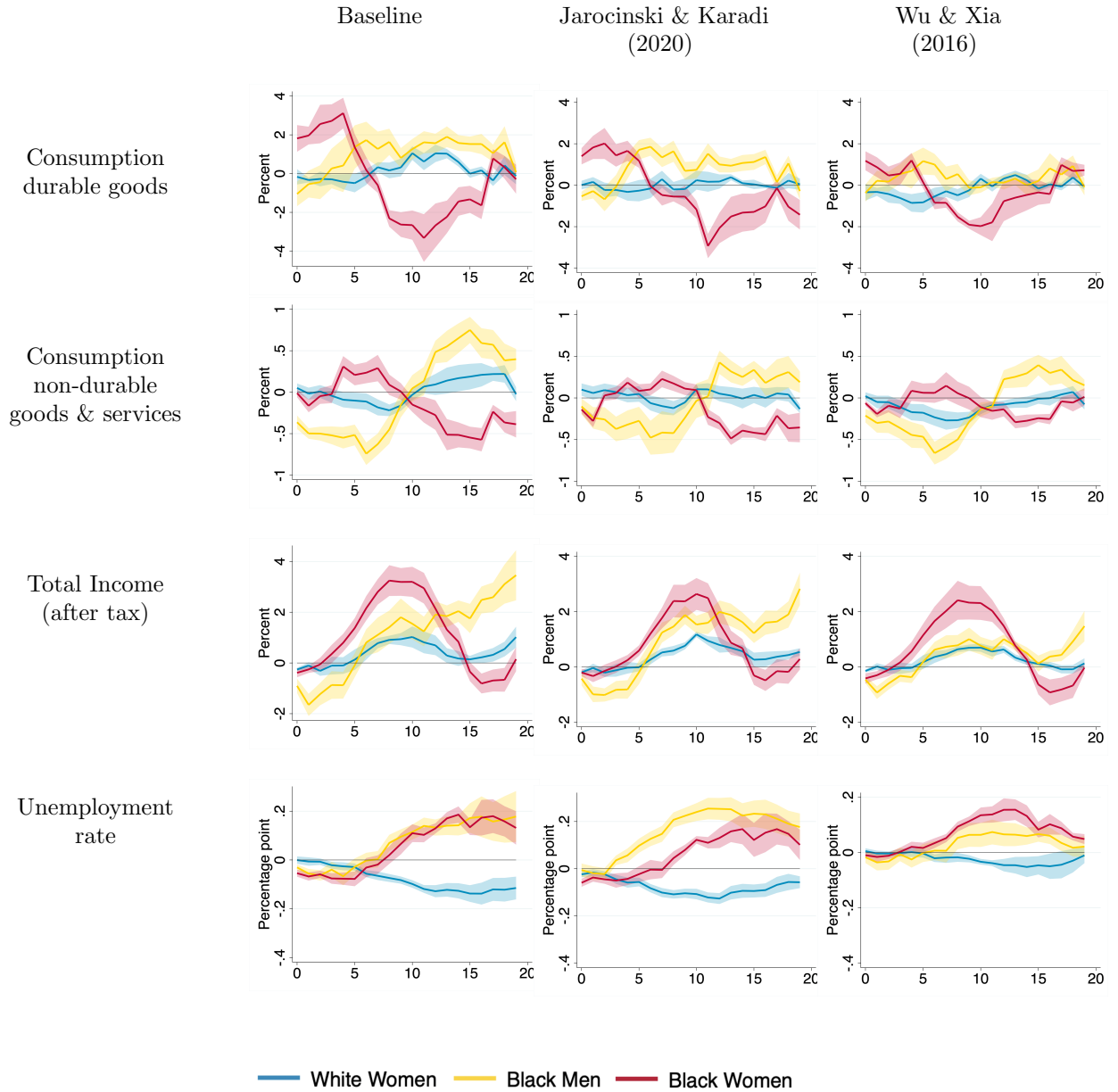
where  $\lambda_t$  is the marginal utility of wealth. When the primary earner faces a negative wage shock ( $\Delta w_1 < 0$ ), the household's marginal utility of wealth  $\lambda$  rises. This induces the spouse to increase labor supply ( $\Delta h_2 > 0$ ), provided  $\phi_2 > 0$ .

**L Appendix: The added income earner ratio to monetary policy shocks at all horizons**

| Quarters | Added Earner Ratio $\xi$ |               |               |              |
|----------|--------------------------|---------------|---------------|--------------|
|          | White men                | White women   | Black men     | Black women  |
| 0-0      | 0.01 (0.02)              | 0.06 (2.70)   | 0.02 (0.02)   | -0.03 (0.05) |
| 0-1      | 0.01 (0.02)              | -0.01 (0.05)  | 0.02 (0.02)*  | -0.00 (0.01) |
| 0-2      | 0.01 (0.01)              | -0.01 (0.05)  | 0.03 (0.02)*  | -0.00 (0.00) |
| 0-3      | 0.01 (0.01)*             | -0.03 (0.13)  | 0.03 (0.03)*  | -0.00 (0.01) |
| 0-4      | 0.01 (0.01)*             | -0.04 (0.16)  | 0.03 (0.03)*  | 0.00 (0.00)  |
| 0-5      | 0.01 (0.01)*             | -0.12 (0.98)  | 0.03 (0.02)*  | 0.00 (0.01)  |
| 0-6      | 0.01 (0.01)*             | -0.23 (3.47)  | 0.04 (0.03)*  | 0.01 (0.01)* |
| 0-7      | 0.01 (0.01)*             | -0.67 (25.30) | 0.05 (0.04)*  | 0.01 (0.01)* |
| 0-8      | 0.01 (0.01)*             | 0.23 (2.41)   | 0.07 (0.07)*  | 0.01 (0.01)* |
| 0-9      | 0.01 (0.01)              | 0.13 (0.72)   | 0.12 (0.18)   | 0.01 (0.01)* |
| 0-10     | 0.01 (0.01)              | 0.11 (0.41)   | 0.16 (0.33)   | 0.01 (0.01)* |
| 0-11     | 0.00 (0.01)              | 0.10 (0.30)   | 0.17 (0.39)   | 0.01 (0.01)* |
| 0-12     | 0.00 (0.01)              | 0.11 (0.31)   | 0.22 (0.73)   | 0.01 (0.01)* |
| 0-13     | -0.00 (0.01)             | 0.11 (0.27)   | 0.53 (4.68)   | 0.01 (0.01)  |
| 0-14     | -0.01 (0.02)             | 0.12 (0.29)   | -0.31 (1.89)  | 0.01 (0.02)  |
| 0-15     | -0.02 (0.03)             | 0.13 (0.31)   | -0.11 (0.25)  | 0.01 (0.01)  |
| 0-16     | -0.03 (0.06)             | 0.16 (0.40)   | -0.04 (0.06)  | -0.00 (0.02) |
| 0-17     | -0.06 (0.12)             | 0.20 (0.56)   | -0.02 (0.03)  | -0.02 (0.06) |
| 0-18     | -0.11 (0.35)             | 0.25 (0.76)   | -0.01 (0.01)* | -0.04 (0.14) |
| 0-19     | -0.29 (2.00)             | 0.29 (0.93)   | -0.01 (0.01)  | -0.05 (0.16) |

Note: The table shows the cumulative percent response to a 25bp contractionary monetary shock at various quarters. Ratio  $\xi$  is calculated as LFP Spouse relative to Consumption Difference. \* Significant at one standard deviation. Newey–West HAC standard errors in parentheses. Data used to calculate the ratio is from both the CE and CPS.

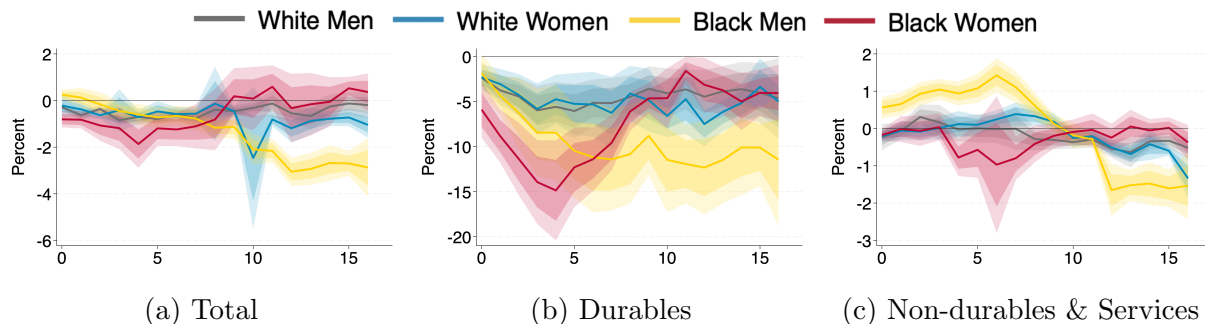
## M Appendix: Effects of alternative monetary policy shocks on gender and racial gaps with white men



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on gaps in outcomes between households headed by white men and white women (blue), black men (yellow), and black women (red). Consumption and income are at the household level from CE data, while the unemployment rate is of individuals from CPS data. The first column has estimates from equation 1. The second and third columns use the Jarocinski and Karadi (2020) monetary policy shock series and the Wu and Xia (2016) shadow rates in place of the instrument and 2-year Treasury yield. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals.

## N Appendix: Effects of monetary policy shocks on household consumption, robustness estimation

Figure N.1: Effect of monetary policy shocks on household consumption with four lags of variable controls



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption, separately by households headed by white men, white women, black men, and black women. Sub-figures show consumption of (a) total goods, (b) durable goods, and (c) non-durable goods and services. Estimates follow equation 1 with four lags instead of two lags of variable controls. Time (horizontal axis) is in quarters. Dark and light shaded areas represent one standard deviation and 90% confidence intervals respectively. Data used is sourced from the CE.

Table N.1: Effect of monetary policy shocks on household consumption using seasonally adjusted series

| Response                   | Group       | Estimate at quarter |                |               |               |
|----------------------------|-------------|---------------------|----------------|---------------|---------------|
|                            |             | 3                   | 6              | 12            | 19            |
| Total                      | White Men   | -1.15 (0.38)*       | -0.48 (0.17)*  | -0.62 (0.58)* | 0.36 (0.27)*  |
|                            | White Women | -0.39 (0.38)*       | -0.01 (0.22)   | -0.47 (0.18)* | -0.49 (0.25)* |
|                            | Black Men   | 1.31 (0.55)*        | 0.41 (0.58)    | -1.73 (0.31)* | -1.62 (0.68)* |
|                            | Black Women | -0.12 (0.33)        | -1.48 (0.71)*  | 1.53 (0.57)*  | -1.16 (0.57)* |
| Durables                   | White Men   | -7.17 (1.89)*       | -6.57 (1.39)*  | -4.16 (0.83)* | -2.61 (1.07)* |
|                            | White Women | -8.22 (1.79)*       | -7.79 (2.28)*  | -5.05 (0.96)* | -2.21 (1.20)* |
|                            | Black Men   | -8.58 (3.14)*       | -6.33 (2.60)*  | -6.44 (1.15)* | -5.74 (2.54)* |
|                            | Black Women | -9.56 (1.68)*       | -13.49 (2.72)* | 2.46 (2.28)*  | -4.36 (1.17)* |
| Non-durables<br>& Services | White Men   | 0.16 (0.23)         | 0.62 (0.30)*   | -0.37 (0.49)  | 0.35 (0.22)*  |
|                            | White Women | 0.74 (0.21)*        | 0.92 (0.13)*   | 0.07 (0.10)   | -0.29 (0.09)* |
|                            | Black Men   | 2.68 (0.28)*        | 1.26 (0.34)*   | -0.73 (0.34)* | -0.72 (0.45)* |
|                            | Black Women | 0.82 (0.29)*        | 0.08 (0.48)    | 1.28 (0.36)*  | -0.66 (0.55)* |

Note: The table shows the effects of a 25bp contractionary monetary policy shock on household consumption at various quarters separately by households headed by white men, white women, black men, and black women. Each household consumption series is seasonally adjusted using the Census X-11 procedure without applying the four-quarter moving average. \* Significant at one standard deviation. Newey–West HAC standard errors in parentheses. Data used is sourced from the CE.

## O Appendix: Economic state-dependent effects of monetary policy shocks on household consumption using smooth state transitions

In this section, I compare estimates of the state-dependent effects of monetary policy shocks on household consumption under alternative methods of identifying states of economic growth and inflation. The baseline state indicator, which in equation 2 classifies observations into high- and low-state periods, is compared to two smooth-transition functions. The first smooth-transition function follows Auerbach and Gorodnichenko (2012) and the second follows the CDF-based approach of Born et al. (2020). These alternative constructions demonstrate that the estimated state dependence in the main text is robust to the method used to characterize each state. The below equation is estimated:

$$x_{g,t+h} = \alpha_{g,h} + \psi_{g,h}(L)Z_{t-1} + \chi_{g,h}^1 Y_t^{2y}(1 - F(z_{t-1})) + \chi_{g,h}^2 Y_t^{2y}F(z_{t-1}) + \varepsilon_{g,t+h} \quad (3)$$

for  $h = 0, \dots, H - 1$ ,

where all variables are specified as in equation 1 and  $0 \leq F(z_{t-1}) \leq 1$  is a smooth-transition state indicator function that determines the weight of each regime (either GDP growth or inflation). The state variable  $z_{t-1}$  is lagged one period so that the macroeconomic conditions used to characterize the state precede the monetary policy shock.

The first state indicator function follows the smooth-transition approach of Auerbach and Gorodnichenko (2012). Each quarterly macroeconomic series, either GDP growth or the inflation rate, is first smoothed using a seven-quarter moving average to reduce the influence of short run fluctuations and is then standardized to become  $z_t$ . The transition function is defined as follows, where  $\gamma = 1.5$  following Auerbach and Gorodnichenko (2012):

$$F(z_t) = \frac{\exp(-\gamma z_t)}{1 + \exp(-\gamma z_t)}, \quad \gamma > 0, \quad \text{var}(z_t) = 1, \quad E(z_t) = 0,$$

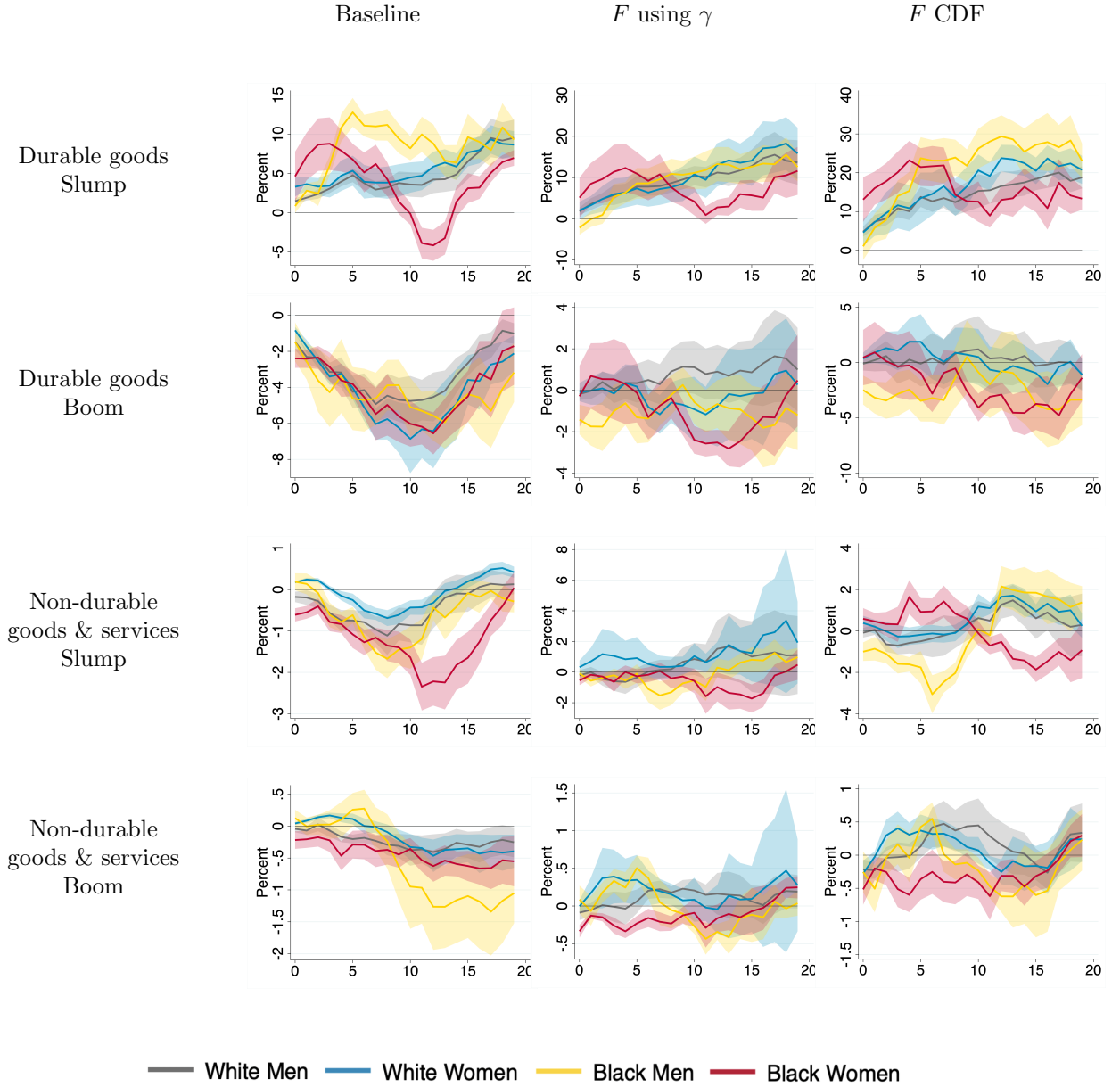
The second approach constructs the state transition indicator using the empirical cumulative distribution function (CDF) of the smoothed state variable following Born et al. (2020). Each macroeconomic series is smoothed using a three-quarter backward-looking moving average.

The baseline state-dependent regressions are estimated by two-stage least squares with

heteroskedasticity-and-autocorrelation-consistent standard errors. First-stage F-statistics are reported separately for the two endogenous regressors in Table O.1. The interaction of the 2-year Treasury yield and  $F(z_{t-1})$  is instrumented with the interaction of the high-frequency monetary policy surprise measure and  $F(z_{t-1})$ . Several first-stage F-statistics fall below the conventional threshold and should be interpreted individually with caution. Similarity of point estimates across alternative state-variable constructions with strong instruments provides evidence regarding the robustness of the results to the method used to characterize macroeconomic conditions, but does not eliminate weak-instrument concerns for individual coefficients.

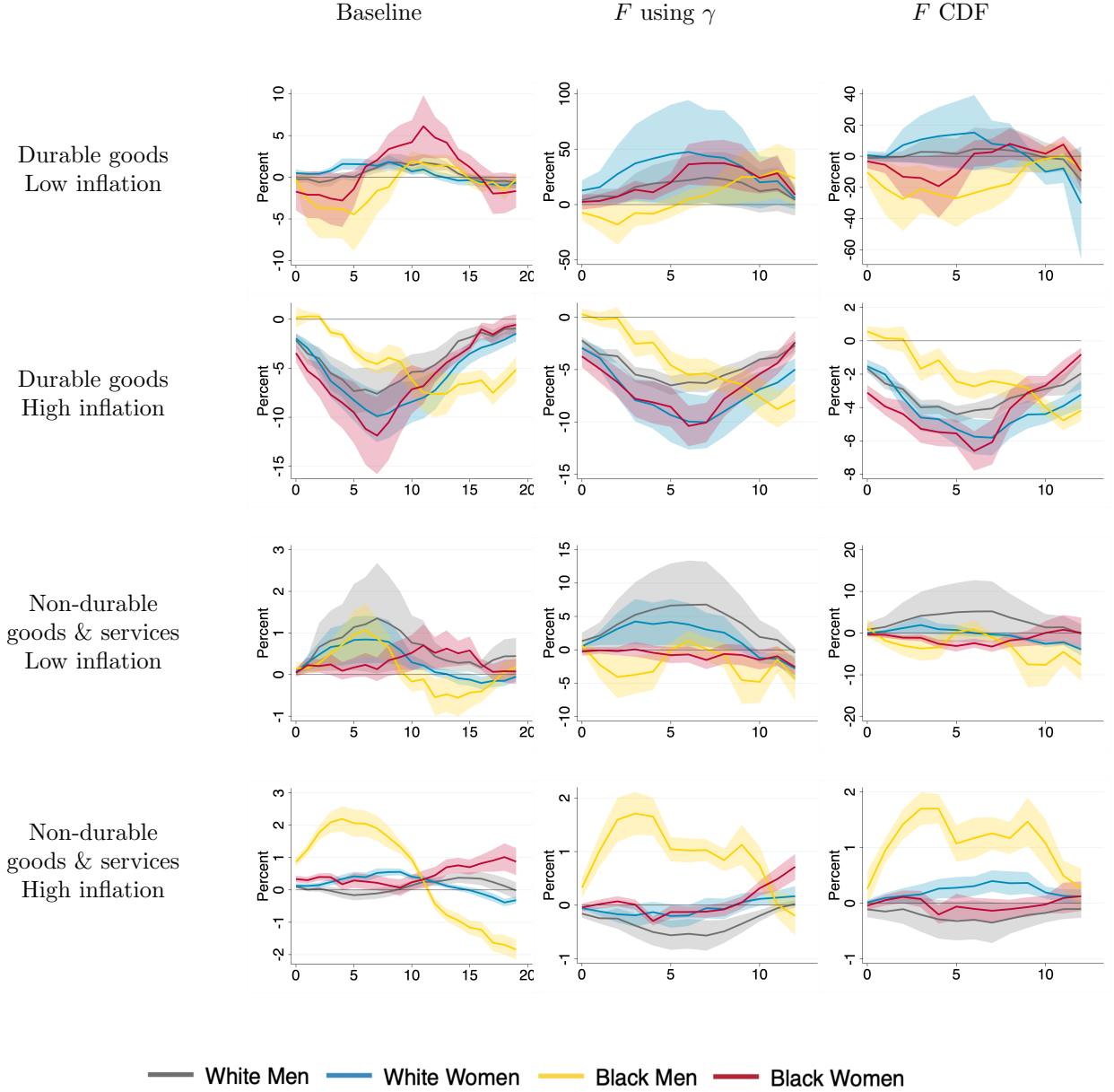
The comparison of estimates for economic states are reported in subsection O.1 and for inflation states in O.2. The inflation-state estimates are reported through horizon 12 since at longer horizons, the estimated standard errors become unstable.

## O.1 Economic states: slump vs boom



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption by economic state, separately by households headed by white men, white women, black men, and black women. The first column has estimates from equation 2. The second and third columns use estimates from equation 3 with the indicator function  $F(z_{t-1})$  containing  $\gamma$  as in Auerbach and Gorodnichenko (2012) and the smoothed CDF as in Born et al. (2020), respectively. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Household data used is sourced from the CE.

## O.2 Inflation states: low vs high



Note: The figure shows the effects of a 25bp contractionary monetary policy shock on household consumption by inflation state, separately by households headed by white men, white women, black men, and black women. The first column has estimates from equation 2. The second and third columns use estimates from equation 3 with the indicator function  $F(z_{t-1})$  containing  $\gamma$  as in Auerbach and Gorodnichenko (2012) and the smoothed CDF as in Born et al. (2020), respectively. Time (horizontal axis) is in quarters. Shaded areas denote one standard deviation confidence intervals. Data used is sourced from the CE.

Table O.1: F-statistics for smooth transition measure of economic and inflation states

| Outcome                 | State           | $F$ Approach | White Men | White Women | Black Men | Black Women |
|-------------------------|-----------------|--------------|-----------|-------------|-----------|-------------|
| Durables                | Slump (Intr)    | Gamma        | 9.66      | 6.53        | 12.26     | 8.33        |
| Durables                | Slump (Intr)    | CDF          | 36.03     | 31.46       | 33.93     | 37.43       |
| Durables                | Boom            | Gamma        | 5.45      | 4.38        | 6.51      | 6.48        |
| Durables                | Boom            | CDF          | 7.72      | 6.23        | 8.44      | 8.67        |
| Durables                | Low infl (Intr) | Gamma        | 4.83      | 12.96       | 15.73     | 11.80       |
| Durables                | Low infl (Intr) | CDF          | 0.47      | 0.92        | 1.30      | 1.28        |
| Durables                | High infl       | Gamma        | 24.92     | 23.38       | 27.16     | 28.00       |
| Durables                | High infl       | CDF          | 16.18     | 11.91       | 17.32     | 18.15       |
| Non-durables & Services | Slump (Intr)    | Gamma        | 3.01      | 3.14        | 7.03      | 4.97        |
| Non-durables & Services | Slump (Intr)    | CDF          | 31.52     | 32.28       | 33.08     | 28.29       |
| Non-durables & Services | Boom            | Gamma        | 4.70      | 5.41        | 7.12      | 4.31        |
| Non-durables & Services | Boom            | CDF          | 8.26      | 7.57        | 8.35      | 5.98        |
| Non-durables & Services | Low infl (Intr) | Gamma        | 7.14      | 6.30        | 16.61     | 1.90        |
| Non-durables & Services | Low infl (Intr) | CDF          | 1.66      | 2.29        | 7.71      | 0.60        |
| Non-durables & Services | High infl       | Gamma        | 27.25     | 29.59       | 61.02     | 21.53       |
| Non-durables & Services | High infl       | CDF          | 16.86     | 22.41       | 39.62     | 14.49       |

Note: The table shows first-stage F-statistics for state-dependent local projections by group and state indicator function approach. The following equation is estimated:  $x_{g,t+h} = \alpha_{g,h} + \psi_{g,h}(L)Z_{t-1} + \beta_{g,h}^e Y_t^{2y} + \beta_{g,h}^F Y_t^{2y} F(z_{t-1}) + \varepsilon_{g,t+h}$  for  $h = 0, \dots, H - 1$ . Each cell is a separate regression, listed in the order of figures in Appendix subsections O.1 and O.2. Groups labeled (Intr) are the first-stage F-statistics of the interaction term between the economic state measure and the monetary policy measure  $\beta^F$ , while other first-stage F-statistics are of  $\beta^e$ . Household data used is sourced from the CE.

## **P Appendix: Effects of monetary policy shocks on household consumption by shock sign**

This section reports estimates of asymmetry in household consumption responses by sign of the monetary policy shock. The responses to contractionary and expansionary shocks are compared both in terms of their signed differences and the magnitude of their responses. The former captures differences in the direction and level of the estimated responses, while the latter tests whether the size of the response is asymmetric across shock types.

Panel A of Table P.1 shows that consumption responses differ significantly by the sign of the monetary policy shock across all horizons. The difference in responses is calculated as the raw difference between the coefficients estimated following contractionary and expansionary shocks.

Panel B of Table P.1 displays estimates of the difference in the absolute values of the responses following contractionary and expansionary shocks. The magnitude of consumption responses significantly vary by shock sign, demonstrating that household consumption responses are asymmetric by shock type. Although household durable goods consumption responds more strongly to expansionary shocks at the immediate horizon, the pattern reverses over the medium and longer terms (quarters 0-12 and 0-19), with all expenditure categories exhibiting larger responses to contractionary shocks. These differences are statistically significant for durable goods consumption among households headed by white men and black women and for non-durable goods and services consumption among households headed by black men.

Table P.1: Difference in household consumption responses by group and shock sign

| Response                                                  | Group       | Difference summed over quarters |                  |                  |
|-----------------------------------------------------------|-------------|---------------------------------|------------------|------------------|
|                                                           |             | Qtr 0-6                         | Qtr 0-12         | Qtr 0-19         |
| <i>Panel A: Difference in responses</i>                   |             |                                 |                  |                  |
| Durables                                                  | White men   | -48.76 (7.30)*                  | -97.35 (12.48)*  | -138.99 (16.04)* |
|                                                           | White Women | -49.50 (8.43)*                  | -125.93 (15.14)* | -197.55 (19.85)* |
|                                                           | Black Men   | -43.02 (10.67)*                 | -108.30 (17.84)* | -197.20 (24.78)* |
|                                                           | Black Women | -91.03 (12.31)*                 | -149.83 (18.49)* | -187.70 (20.69)* |
| Non-durables<br>& Services                                | White men   | 3.38 (1.20)*                    | 9.12 (2.04)*     | 13.43 (2.78)*    |
|                                                           | White Women | 5.03 (0.97)*                    | 10.90 (1.56)*    | 9.05 (2.09)*     |
|                                                           | Black Men   | 16.89 (2.35)*                   | 25.69 (3.16)*    | 14.19 (3.86)*    |
|                                                           | Black Women | 4.54 (1.53)*                    | 10.73 (2.67)*    | 21.27 (3.90)*    |
| <i>Panel B: Difference in absolute value of responses</i> |             |                                 |                  |                  |
| Durables                                                  | White men   | 1.86 (7.62)                     | 19.96 (13.47)*   | 20.82 (17.57)*   |
|                                                           | White Women | -9.87 (8.43)*                   | 0.38 (14.72)     | 2.74 (19.51)     |
|                                                           | Black Men   | -15.44 (11.36)*                 | -19.38 (20.01)   | -24.17 (25.54)   |
|                                                           | Black Women | -13.69 (11.95)*                 | 20.39 (17.91)*   | 20.32 (20.64)    |
| Non-durables<br>& Services                                | White men   | -0.28 (1.19)                    | -0.37 (2.06)     | 1.28 (2.65)      |
|                                                           | White Women | 0.65 (1.06)                     | 0.41 (1.66)      | -0.05 (2.08)     |
|                                                           | Black Men   | 2.13 (1.98)*                    | 2.84 (2.57)*     | 5.59 (3.17)*     |
|                                                           | Black Women | -0.32 (1.34)                    | -1.75 (2.26)     | -2.88 (3.29)     |

Note: The table shows the difference (panel A) and difference of the absolute value (panel B) in area under the impulse response of each household in Figure 8c-d. Differences are calculated at three time trajectories; from quarter 0-6, 0-12, and 0-19. \* Significant at one standard deviation. Standard error in parentheses. Household data used is sourced from the CE.

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